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Quality Assurance (IM353IA): Unit II

Department of Industrial Engineering and Management

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Unit II

Control Charts for Variables and Attributes | Process Capability

Control Charts for Variables

Chapter 5. Control Charts for Variables

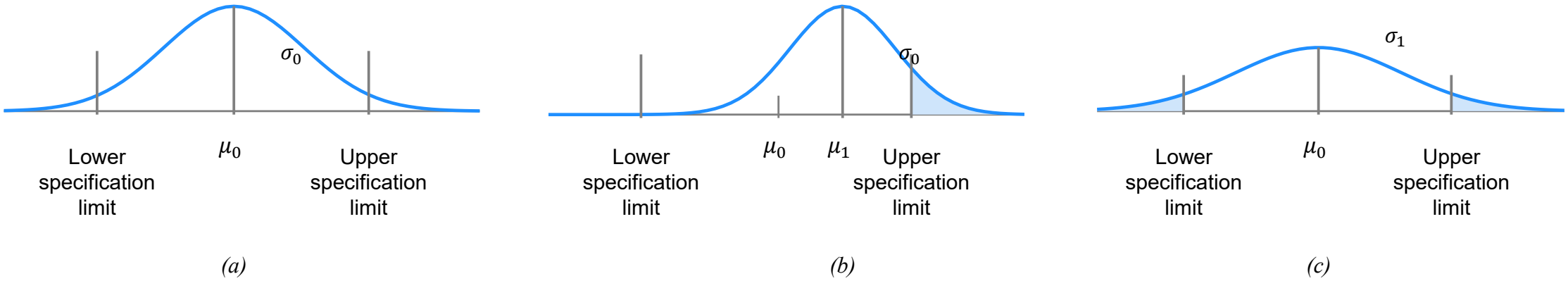


Figure 5-1 The need for controlling both process mean and process variability. (*a*) Mean and standard deviation at nominal levels. (*b*) Process mean $\mu_1 > \mu_0$. (*c*) Process standard deviation $\sigma_1 > \sigma_0$.

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Control Charts for $\overline{x}$ and R

x: quantity of interest

$x_1, x_2, \ldots, x_n$: samples of x

$$x \sim N(\mu, \sigma)$$

$$\overline{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$\overline{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$\mu + Z_{\alpha/2}\sigma_{\overline{x}} = \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \mu - Z_{\alpha/2}\sigma_{\overline{x}} = \mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \tag{5-1}$$

Subgroup Data with Unknown μ and σ

$\overline{\overline{x}}$: grand average of x, best estimate for μ

$$\overline{\overline{x}} = \frac{\overline{x}_1 + \overline{x}_2 + \cdots + \overline{x}_m}{m} \tag{5-2}$$

$$R = x_{\max} - x_{\min}$$

$R_1, R_2, \ldots, R_m$: ranges of m samples

$$\overline{R} = \frac{R_1 + R_2 + \cdots + R_m}{m} \tag{5-3}$$

Control Limits for the $\overline{x}$ Chart

$$\text{UCL} = \overline{\overline{x}} + A_2\overline{R}$$
$$\text{Center line} = \overline{\overline{x}}$$
$$\text{LCL} = \overline{\overline{x}} - A_2\overline{R} \quad (5\text{-}4)$$

The constant A_2 is tabulated for various sample sizes in Appendix Table VI.

Control Limits for the R Chart

$$\text{UCL} = D_4\overline{R}$$
$$\text{Center line} = \overline{R}$$
$$\text{LCL} = D_3\overline{R} \quad (5\text{-}5)$$

The constants D_3 and D_4 are tabulated for various values of n in Appendix Table VI.

Phase I Application of $\bar{x}$ and R Charts

- Equations 5-4 and 5-5 are trial control limits.
  - Determined from m initial samples.
    - Typically 20-25 subgroups of size n between 3 and 5.
  - Any out-of-control points should be examined for assignable causes.
    - If assignable causes are found, discard points from calculations and revise the trial control limits.
    - Continue examination until all points plot in control.
    - Adopt resulting trial control limits for use.
    - If no assignable cause is found, there are two options.
      1. Eliminate point as if an assignable cause were found and revise limits.
      2. Retain point and consider limits appropriate for control.
  - If there are many out-of-control points they should be examined for patterns that may identify underlying process problems.

Example 5-1

Table 5-1 Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	1	2	3	4	5	$\bar{x}_i$	R_i
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470

Example 5-1 (cont'd)

Table 5-1 Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	1	2	3	4	5	$\bar{x}_i$	R_i
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224
					$\Sigma\bar{x}_i = 37.6400$		$\Sigma R_i = 8.1302$
					$\bar{\bar{x}} = 1.5056$		$\bar{R} = 0.32521$

$$\overline{R} = \frac{\sum_{i=1}^{25} R_i}{25} = \frac{8.1302}{25} = 0.32521$$

$$\text{LCL} = \overline{R}D_3 = 0.32521(0) = 0$$
$$\text{UCL} = \overline{R}D_4 = 0.32521(2.114) = 0.68749$$

$$\overline{\overline{x}} = \frac{\sum_{i=1}^{25} \overline{x}_i}{25} = \frac{37.6400}{25} = 1.5056$$

$$\text{UCL} = \overline{\overline{x}} + A_2\overline{R} = 1.5056 + (0.577)(0.32521) = 1.69325$$
$$\text{LCL} = \overline{\overline{x}} - A_2\overline{R} = 1.5056 - (0.577)(0.32521) = 1.31795$$

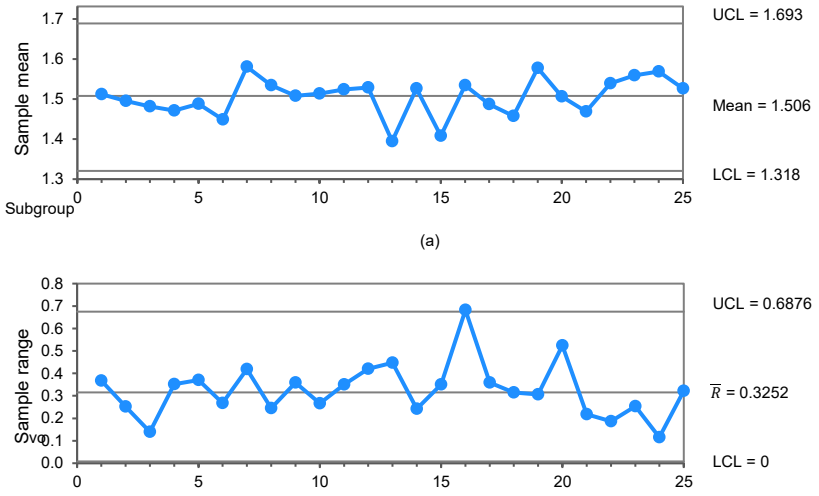


Figure 5-2 $\bar{x}$ and R charts (from Minitab) for flow width in the hard-bake process.

$$\hat{\sigma} = \frac{\overline{R}}{d_2} = \frac{0.32521}{2.326} = 0.1398 \text{ microns}$$

Assume spec tolerance is 1.5 +/- 0.5 micron.
Nonconformance probability:

$$p = P\{x < 1.00\} + P\{x > 2.00\}$$
$$= \Phi\left(\frac{1.00 - 1.5056}{0.1398}\right) + 1 - \Phi\left(\frac{2.00 - 1.5056}{0.1398}\right)$$
$$= \Phi(-3.61660) + 1 - \Phi(3.53648)$$
$$\simeq 0.00015 + 1 - 0.99980$$
$$\simeq 0.00035$$

C_p: Process Capability Ration (PCR)

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \tag{5-11}$$

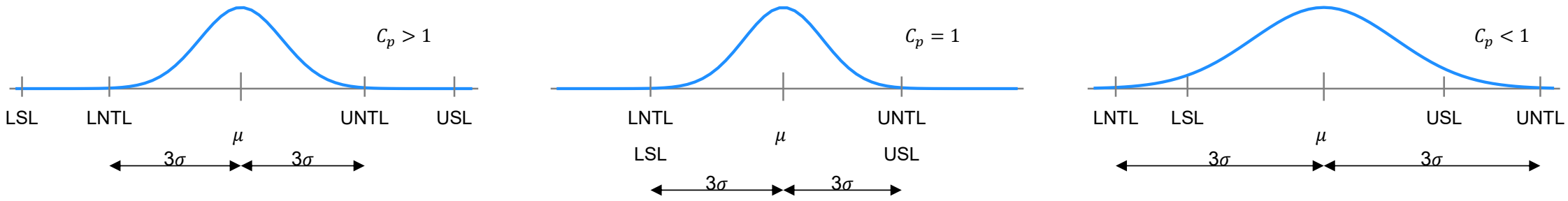
Note: 6σ spread is the basic definition of process capability. 3σ above mean and 3σ below.
If σ is unknown, we can use $\hat{\sigma} = \overline{R}/d_2$. $\hat{\sigma}$ in the example is 0.1398.

$$\hat{C}_p = \frac{2.00 - 1.00}{6(0.1398)} = \frac{1.00}{0.8388} = 1.192$$

$$P = \left(\frac{1}{C_p}\right)100\%$$

P: % of specification band the process uses up. P can be estimated as:

$$\hat{P} = \left(\frac{1}{\hat{C}_p}\right)100\% = \left(\frac{1}{1.192}\right)100\% = 83.89$$



(a) *(b)* *(c)*

Figure 5-3 Process fallout and the process capability ratio C_p.

Revision of Control Limits and Center Lines

- Effective use of control charts requires periodic review and revision of control limits and center lines.
- Sometimes users replace the center line on the $\overline{x}$ chart with a target value.
- When R chart is out of control, out-of-control points are often eliminated to re-compute a revised value of $\overline{R}$ which is used to determine new limits and center line on R chart and new limits on $\overline{x}$ chart.

Phase II Operation of Charts

- Use of control chart for monitoring future production, after a set of reliable limits are established, is called *phase II* of control chart usage (Figure 5-4).
- A run chart showing individuals observations in each sample, called a tolerance chart or tier diagram (Figure 5-5), may reveal patterns or unusual observations in the data.

Example 5-1 (cont'd)

Table 5-2 Additional Samples for Example 5-1

Sample Number	1	2	3	4	5	$\bar{x}_i$	R_i
26	1.4483	1.5458	1.4538	1.4303	1.6206	1.4998	0.1903
27	1.5435	1.6899	1.5830	1.3358	1.4187	1.5142	0.3541
28	1.5175	1.3446	1.4723	1.6657	1.6661	1.5332	0.3215
29	1.5454	1.0931	1.4072	1.5039	1.5264	1.4152	0.4523
30	1.4418	1.5059	1.5124	1.4620	1.6263	1.5097	0.1845
31	1.4301	1.2725	1.5945	1.5397	1.5252	1.4724	0.3220
32	1.4981	1.4506	1.6174	1.5837	1.4962	1.5292	0.1668
33	1.3009	1.5060	1.6231	1.5831	1.6454	1.5317	0.3445
34	1.4132	1.4603	1.5808	1.7111	1.7313	1.5793	0.3181
35	1.3817	1.3135	1.4953	1.4894	1.4596	1.4279	0.1818
36	1.5765	1.7014	1.4026	1.2773	1.4541	1.4824	0.4241
37	1.4936	1.4373	1.5139	1.4808	1.5293	1.4910	0.0920
38	1.5729	1.6738	1.5048	1.5651	1.7473	1.6128	0.2425
39	1.8089	1.5513	1.8250	1.4389	1.6558	1.6560	0.3861
40	1.6236	1.5393	1.6738	1.8698	1.5036	1.6420	0.3662
41	1.4120	1.7931	1.7345	1.6391	1.7791	1.6716	0.3811
42	1.7372	1.5663	1.4910	1.7809	1.5504	1.6252	0.2899
43	1.5971	1.7394	1.6832	1.6677	1.7974	1.6970	0.2003
44	1.4295	1.6536	1.9134	1.7272	1.4370	1.6321	0.4839
45	1.6217	1.8220	1.7915	1.6744	1.9404	1.7700	0.3187

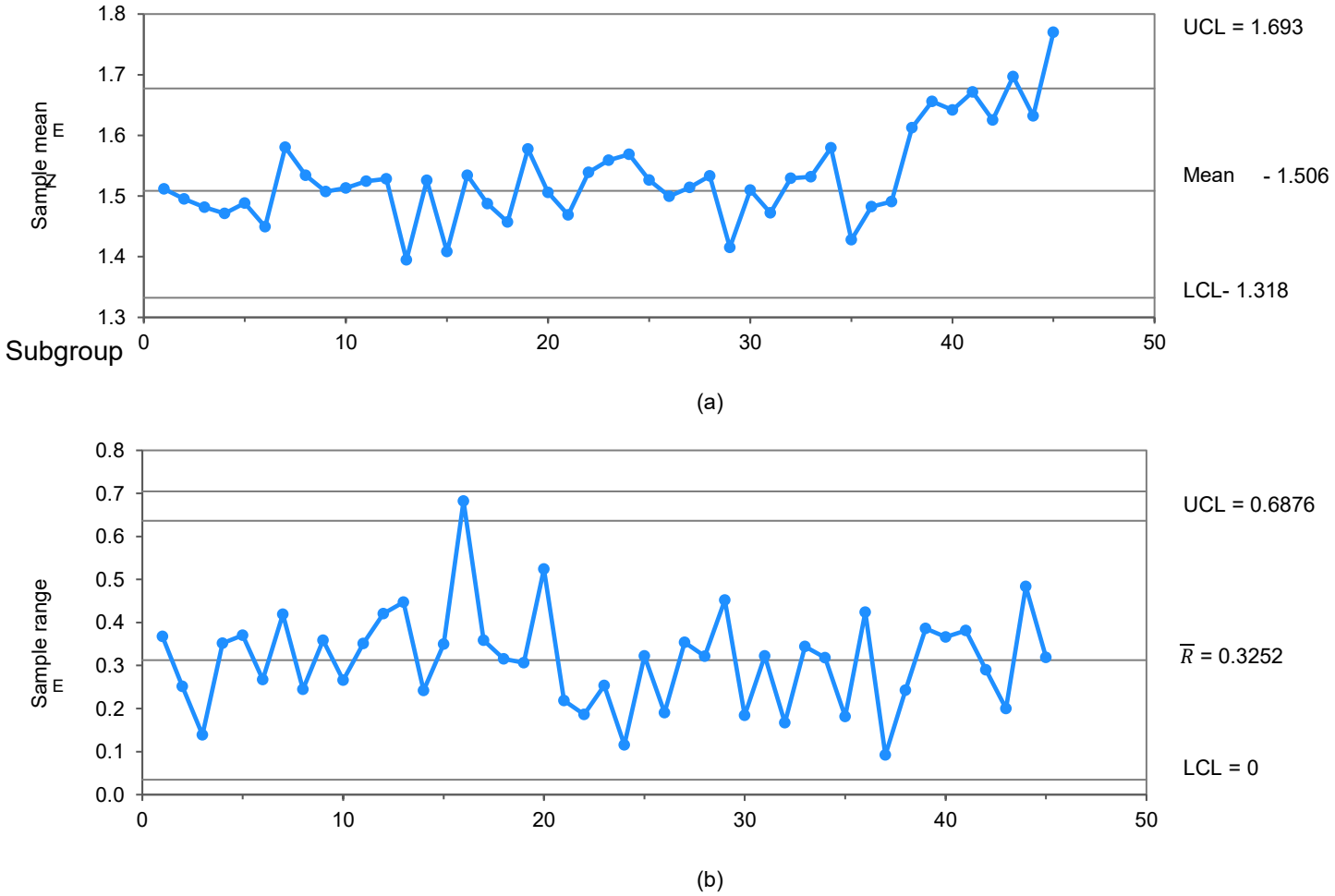


Figure 5-4 Continuation of the $\bar{x}$ and R charts in Example 5-1.

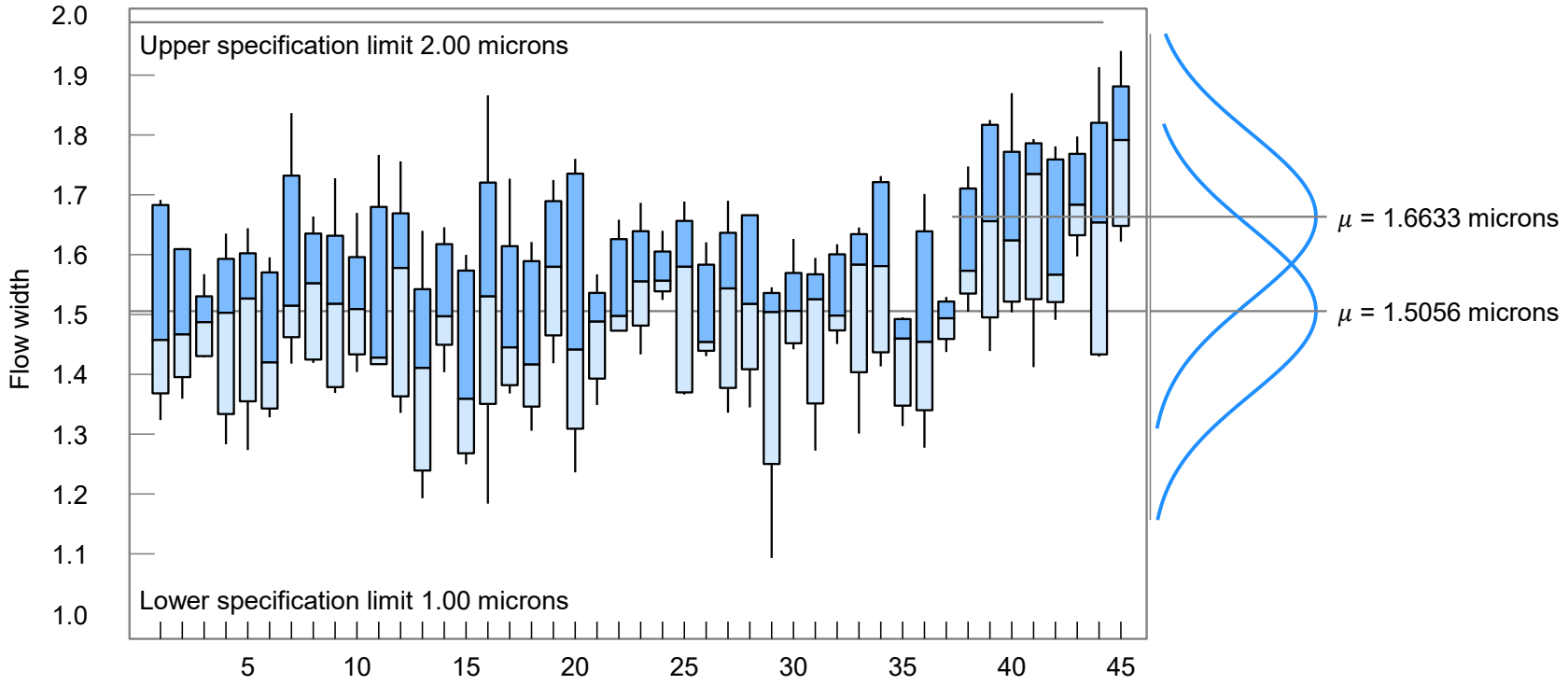


Figure 5-5 Tier chart constructed using the Minitab box plot procedure for the flow width data.

Control vs. Specification Limits

- Control limits are derived from natural process variability, or the natural tolerance limits of a process.
- Specification limits are determined externally, for example by customers or designers.
- There is no mathematical or statistical relationship between the control limits and the specification limits.

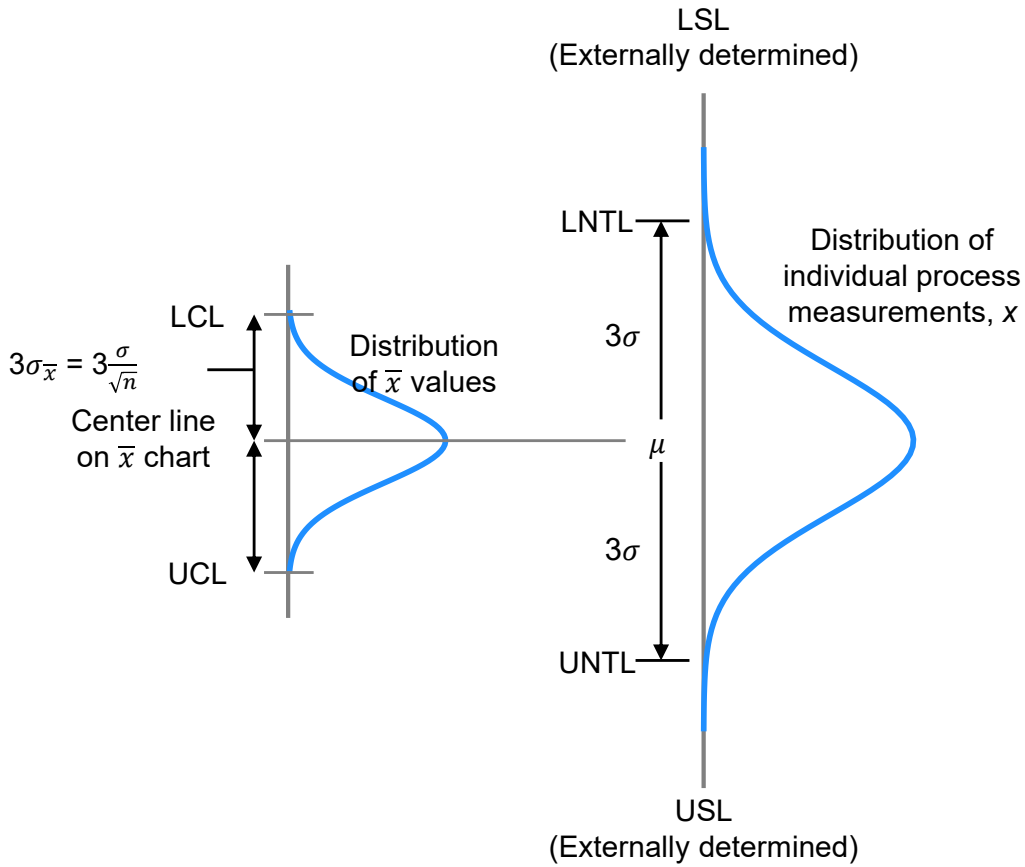


Figure 5-6 Relationship of natural tolerance limits, control limits, and specification limits.

Rational Subgroups

- $\overline{x}$ charts monitor between-sample variability.
- R charts measure within-sample variability.
- Standard deviation estimate of σ used to construct control limits is calculated from within-sample variability.

- It is not correct to estimate σ using

$$s = \sqrt{\frac{\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \overline{\overline{x}}\right)^2}{mn-1}}$$

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Guidelines for Control Chart Design

- Control chart design requires specification of sample size, control limit width, and sampling frequency.
  - Exact solution requires detailed information on statistical characteristics as well as economic factors.
  - The problem of choosing sample size and sampling frequency is one of allocating sampling effort.
- For $\bar{x}$ chart, choose as small a sample size consistent with magnitude of process shift one is trying to detect. For moderate to large shifts, relatively small samples are effective. For small shifts, larger samples are needed.
- For small samples, R chart is relatively insensitive to changes in process standard deviation. For larger samples (n > 10 or 12), s or s^2 charts are better choices.
- NOTE: Skip Section on Changing Sample Size (pages 209-212)

Charts Based on Standard Values

$$\text{UCL} = \mu + 3\frac{\sigma}{\sqrt{n}}$$
$$\text{Center line} = \mu$$
$$\text{LCL} = \mu - 3\frac{\sigma}{\sqrt{n}} \qquad (5\text{-}14)$$

$$\text{UCL} = \mu + A\sigma$$
$$\text{Center line} = \mu$$
$$\text{LCL} = \mu - A\sigma \qquad (5\text{-}15)$$

$$\text{UCL} = D_2\sigma$$
$$\text{Center line} = d_2\sigma$$
$$\text{LCL} = D_1\sigma \qquad (5\text{-}17)$$

$D_1 = d_2 - 3d_3$
$D_2 = d_2 + 3d_3$

d_2 : mean of distribution of relative range
d_3 : standard deviation of distribution of relative range

Interpretation of $\overline{x}$ and R Charts

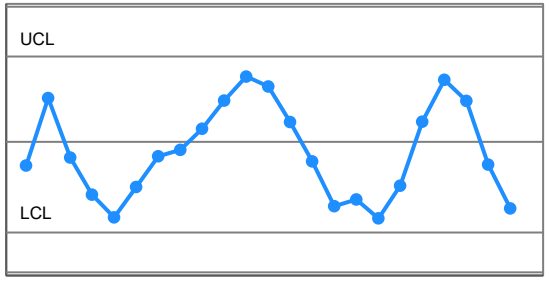


Figure 5-8 Cycles on a control chart.

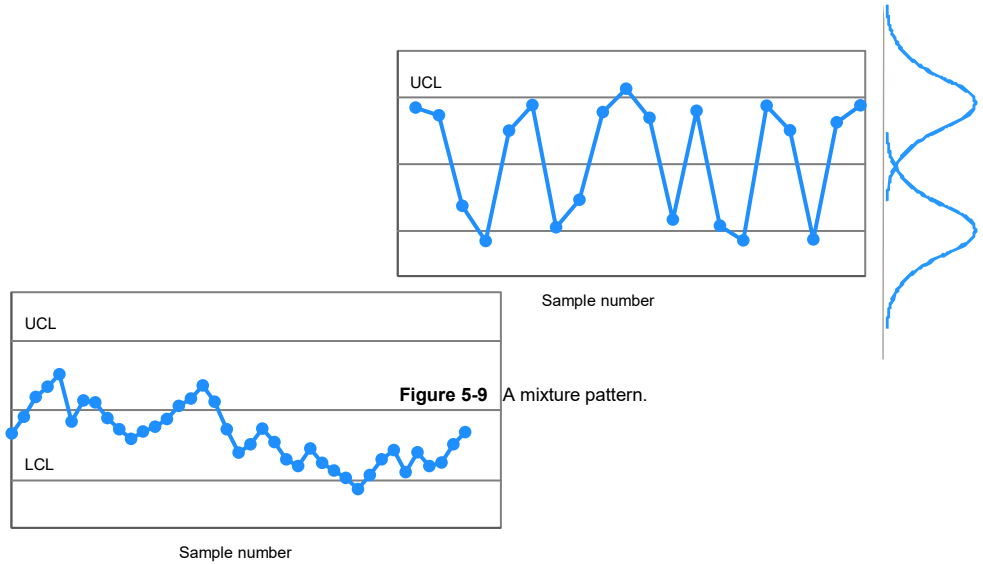


Figure 5-9 A mixture pattern.

Figure 5-10 A shift in process level.

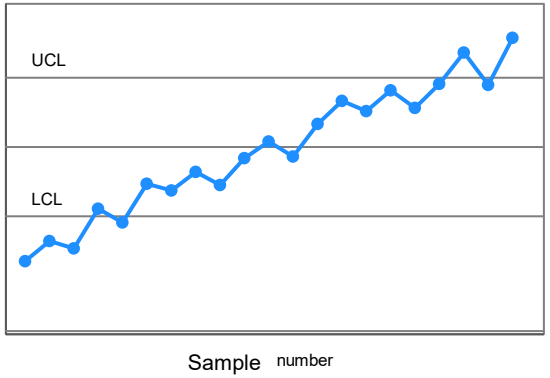


Figure 5-11 A trend in process level.

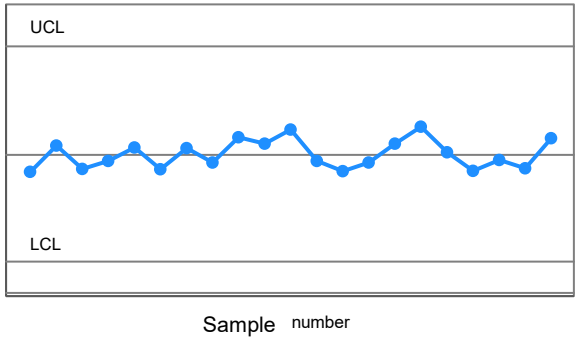


Figure 5-12 A stratification pattern.

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Effect of Nonnormality on $\bar{x}$ and R Charts

- An assumption in performance properties is that the underlying distribution of quality characteristic is normal.
  - If underlying distribution is not normal, sampling distributions can be derived and exact probability limits obtained.
- Usual normal theory control limits are very robust to normality assumption.
- In most cases, samples of size 4 or 5 are sufficient to ensure reasonable robustness to normality assumption for $\bar{x}$ chart.
- Sampling distribution of R is not symmetric, thus symmetric 3-sigma limits are an approximation and α-risk is not 0.0027. R chart is more sensitive to departures from normality than $\bar{x}$ chart.
- Assumptions of normality and independence are not a primary concern in Phase I.

Operating Characteristic (OC) Function

σ is known. In-control mean: μ_0 out of control mean: $\mu_1 = \mu_0 + k\sigma$
Probability of not detecting shift: β-risk
L: number of σ's

$$\beta = \Phi\left(L - k\sqrt{n}\right) - \Phi\left(-L - k\sqrt{n}\right) \quad (5\text{-}19)$$

For $L = 3$, $n = 5$, $k = 2$.

$$\beta = \Phi\left[3 - 2\sqrt{5}\right] - \Phi\left[-3 - 2\sqrt{5}\right]$$
$$= \Phi(-1.47) - \Phi(-7.37)$$
$$\cong 0.0708$$

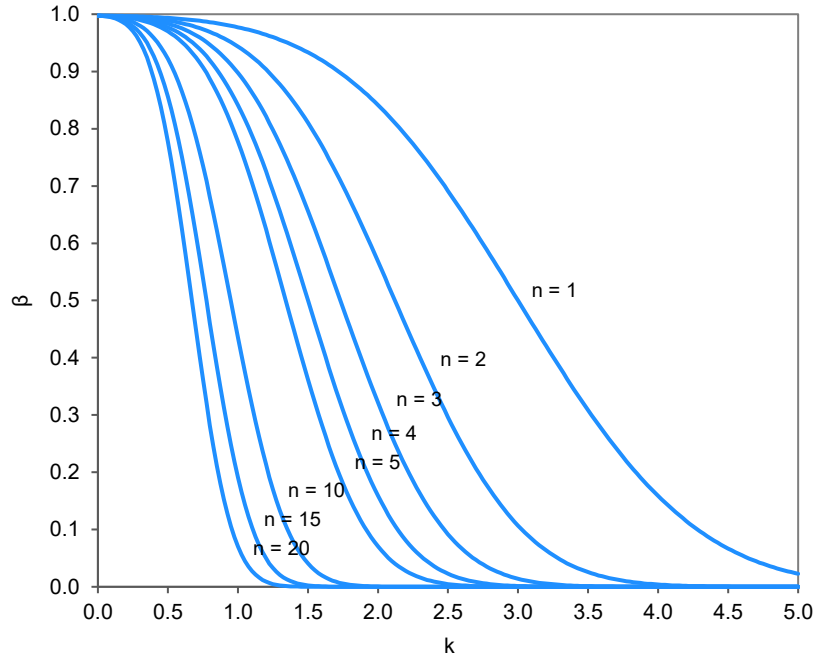


Figure 5-13 Operating-characteristic curves for the $\bar{x}$ chart with three-sigma limits. $\beta = P$ (not detecting a shift of $k\sigma$ in the mean on the first sample following the shift).

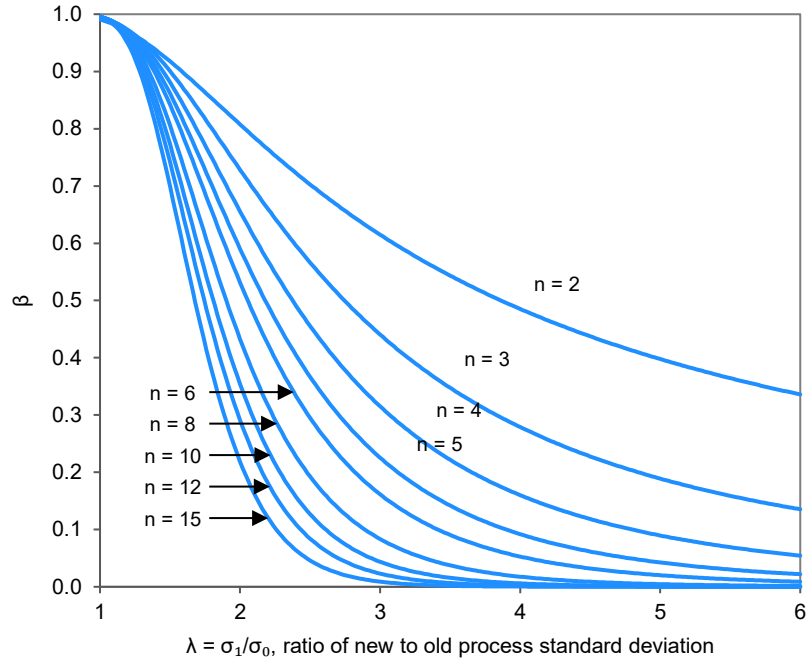


Figure 5-14 Operating-characteristic curves for the *R* chart with three-sigma limits. (Adapted from A. J. Duncan, "Operating Characteristics of *R* Charts," *Industrial Quality Control*, vol. 7, no. 5, pp. 40–41, 1951, with permission of the American Society for Quality Control.)

Average run length (r): shift is detected in the rth sample.

$$\text{ARL} = \sum_{r=1}^{\infty} r\,\beta^{r-1}(1-\beta) = \frac{1}{1-\beta}$$

In the example.

$$\text{ARL} = \frac{1}{1-\beta} = \frac{1}{0.25} = 4$$

Expected number of samples for detecting shift = 4.

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Average Run Length for $\overline{x}$ Chart

For Shewhart control chart:

$$\text{ARL} = \frac{1}{P(\text{one point plots out of control})}$$

$$\text{ARL}_0 = \frac{1}{\alpha} \quad (5\text{-}20)$$

$$\text{ARL}_1 = \frac{1}{1-\beta} \quad (5\text{-}21)$$

Average time to signal (ATS)

$$\text{ATS} = \text{ARL}\, h \quad (5\text{-}22)$$

Average number of individual units sampled for detection (I)

$$I = n\,\text{ARL} \quad (5\text{-}23)$$

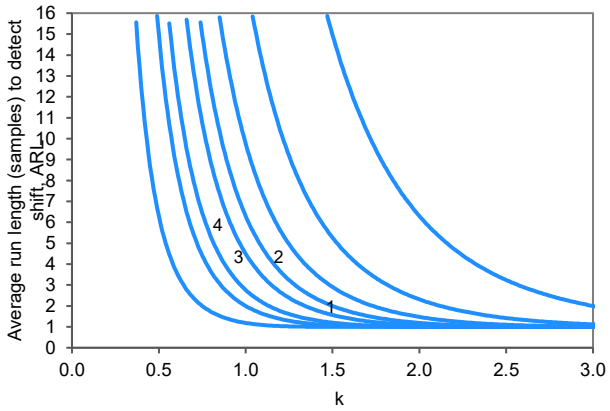


Figure 5-15 Average run length (samples) for the $\overline{x}$ chart with 3-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

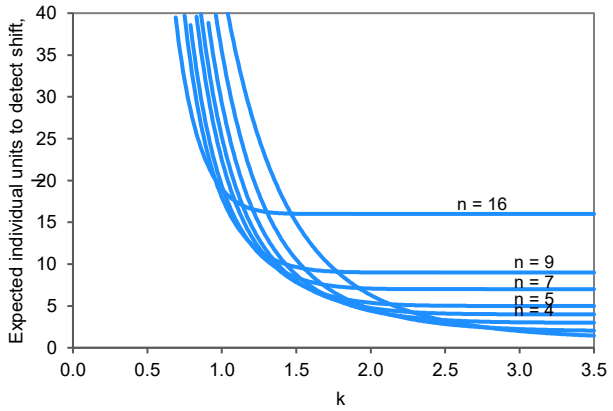


Figure 5-16 Average run length (individual units) for the $\overline{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

Figure 5-15 Average run length (samples) for the $\overline{x}$ chart with 3-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

Figure 5-16 Average run length (individual units) for the $\overline{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

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Control Charts for $\bar{x}$ and s

Use the $\bar{x}$ and s charts instead of the $\bar{x}$ and R charts when:

1. The sample size n is moderately large—say, $n > 10$ or 12. (Recall that the range method for estimating σ loses statistical efficiency for moderate to large samples.)
2. The sample size n is variable.

$$\begin{aligned} \text{UCL} &= B_6\sigma \\ \text{Center line} &= c_4\sigma \\ \text{LCL} &= B_5\sigma \end{aligned} \qquad (5\text{-}25)$$

$$B_5 = c_4 - 3\sqrt{1 - c_4^2} \qquad \text{and} \qquad B_6 = c_4 + 3\sqrt{1 - c_4^2}$$

Assume no standard is given for σ. ⇒ Need to estimate.
m preliminary samples, each of size n.
s_i : standard deviation for i^{th} sample

$$\bar{s} = \frac{1}{m}\sum_{i=1}^{m} s_i$$

$\frac{\bar{s}}{c_4}$: unbiased estimator for σ

s chart has the following parameters:

$$\text{UCL} = \bar{s} + 3\frac{\bar{s}}{c_4}\sqrt{1 - c_4^2}$$

$$\text{Center line} = \bar{s}$$

$$\text{LCL} = \bar{s} - 3\frac{\bar{s}}{c_4}\sqrt{1 - c_4^2}$$

Note: $B_3 = 1 - \frac{3}{c_4}\sqrt{1 - c_4^2}$ and $B_4 = 1 + \frac{3}{c_4}\sqrt{1 - c_4^2}$ Then:

$$\text{UCL} = B_4\bar{s}$$
$$\text{Center line} = \bar{s}$$
$$\text{LCL} = B_3\bar{s}$$

(5-27)

When $\frac{\overline{s}}{c_4}$ is used to estimate σ, $\overline{x}$ chart has the following parameters:

$$\text{UCL} = \overline{\overline{x}} + \frac{3\overline{s}}{c_4\sqrt{n}}$$
$$\text{Center line} = \overline{\overline{x}}$$
$$\text{LCL} = \overline{\overline{x}} - \frac{3\overline{s}}{c_4\sqrt{n}}$$

Define $A_3 = \frac{3}{c_4\sqrt{n}}$. Then:

$$\text{UCL} = \overline{\overline{x}} + A_3\overline{s}$$
$$\text{Center line} = \overline{\overline{x}}$$
$$\text{LCL} = \overline{\overline{x}} - A_3\overline{s}$$

(5-28)

Example 5-3

Table 5-3 Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001	74.011	74.004	74.001	0.0075
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985	73.993	73.996	0.0087
7	73.995	74.006	73.994	74.000	74.005	74.000	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005	74.004	74.004	0.0055
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998	73.997	74.012	73.998	0.0105

Example 5-3 (cont'd)

Table 5-3 Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998	73.999	74.007	74.006	0.0073
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005	74.007	74.001	0.0106
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013	74.020	74.003	74.009	0.0080
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162
					$\Sigma = 1850.028$		0.2351
					$\bar{x} = 74.001$		$\bar{s} = 0.0094$

$$\overline{\overline{x}} = \frac{1}{25}\sum_{i=1}^{25}\overline{x}_i = \frac{1}{25}(1850.028) = 74.001$$

$$\overline{s} = \frac{1}{25}\sum_{i=1}^{25} s_i = \frac{1}{25}(0.2351) = 0.0094$$

For $\overline{x}$ chart:

$$\text{UCL} = \overline{\overline{x}} + A_3\overline{s} = 74.001 + (1.427)(0.0094) = 74.014$$
$$\text{CL} = \overline{\overline{x}} = 74.001$$
$$\text{LCL} = \overline{\overline{x}} - A_3\overline{s} = 74.001 - (1.427)(0.0094) = 73.988$$

For s chart:

$$\text{UCL} = B_4\overline{s} = (2.089)(0.0094) = 0.0196$$
$$\text{CL} = \overline{s} = 0.0094$$
$$\text{LCL} = B_3\overline{s} = (0)(0.0094) = 0$$



Figure 5-17 The $\overline{x}$ and s control charts for Example 5-3. (*a*) The $\overline{x}$ chart with control limits based on $\overline{s}$. (*b*) The s control chart.

$\bar{x}$ and s Control Charts with Variable Sample Size

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{m} n_i \bar{x}_i}{\sum_{i=1}^{m} n_i} \tag{5-30}$$

$$\bar{s} = \left[\frac{\sum_{i=1}^{m} (n_i - 1) s_i^2}{\sum_{i=1}^{m} n_i - m} \right]^{1/2} \tag{5-31}$$

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Example 5-4

Table 5-4 Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001			73.996	0.0046
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985		73.996	0.0099
7	73.995	74.006	73.994	74.000		73.999	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005		74.004	0.0064
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998			73.994	0.0100

Example 5-4 (cont'd)

Table 5-4 Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998			74.008	0.0087
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005		73.999	0.0115
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013			74.008	0.0068
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162

$$\overline{\overline{x}} = \frac{\sum_{i=1}^{25} n_i \overline{x}_i}{\sum_{i=1}^{25} n_i} = \frac{5(74.010) + 3(73.996) + \cdots + 5(73.998)}{5 + 3 + \cdots + 5}$$

$$= \frac{8362.075}{113} = 74.001$$

$$\overline{s} = \left[\frac{\sum_{i=1}^{25}(n_i - 1)s_i^2}{\sum_{i=1}^{25} n_i - 25}\right]^{1/2} = \left[\frac{4(0.0148)^2 + 2(0.0046)^2 + \cdots + 4(0.0162)^2}{5 + 3 + \cdots + 5 - 25}\right]^{1/2}$$

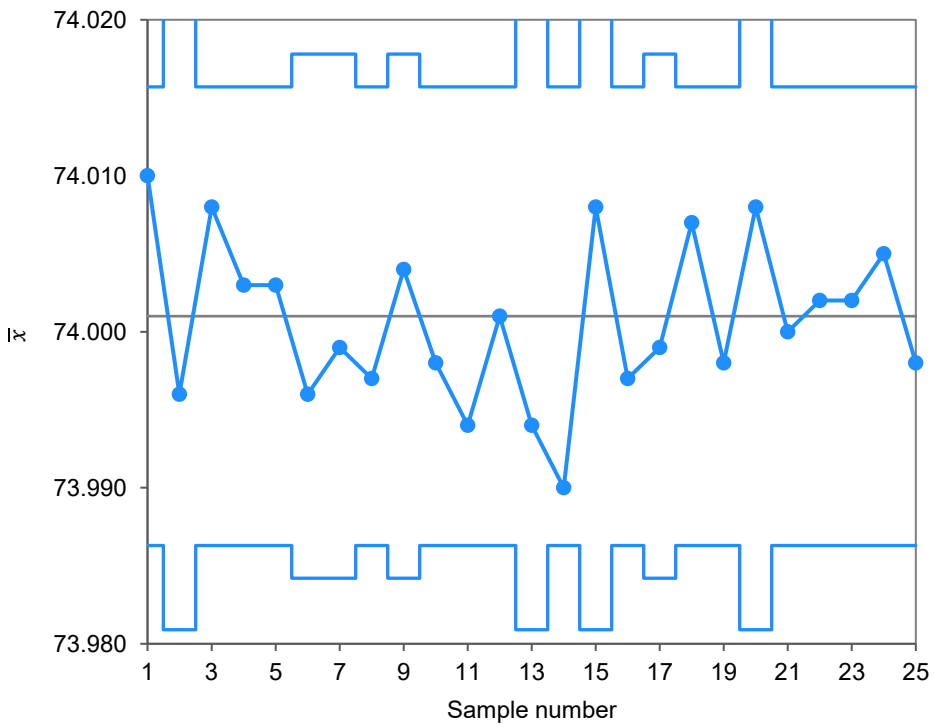
$$= \left[\frac{0.009324}{88}\right]^{1/2} = 0.0103$$

For $\overline{x}$ chart:

$\text{UCL} = 74.001 + (1.427)(0.0103) = 74.016$
$\text{CL} = 74.001$
$\text{LCL} = 74.001 - (1.427)(0.0103) = 73.986$

For s chart:

$\text{UCL} = (2.089)(0.0103) = 0.022$
$\text{CL} = 0.0098$
$\text{LCL} = 0(0.0103) = 0$



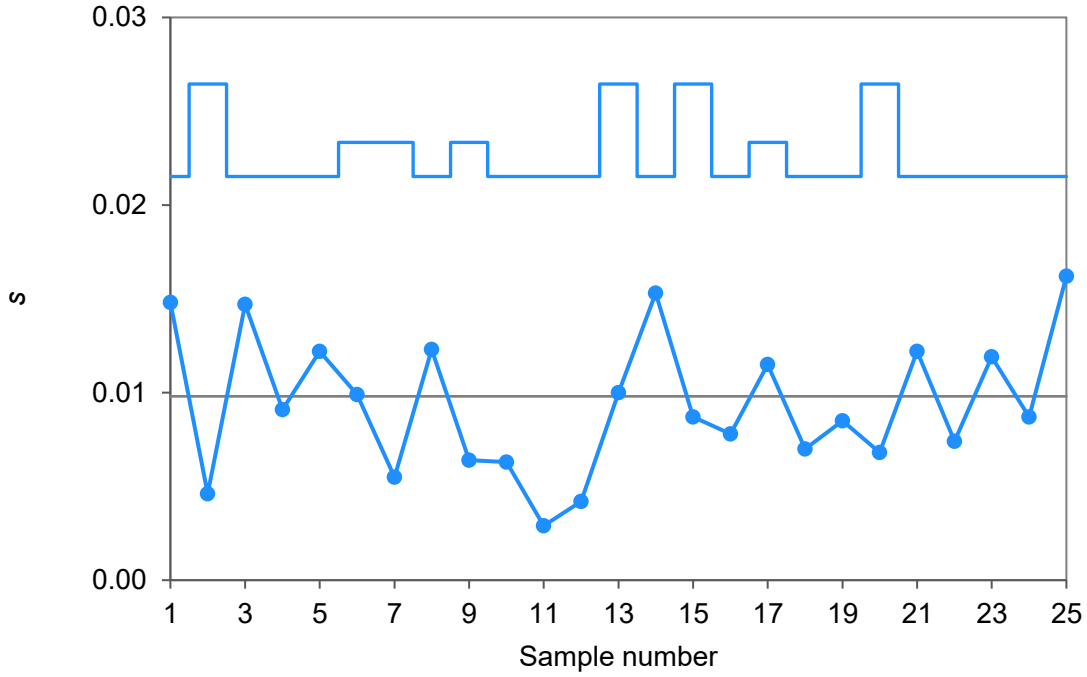


Figure 5-18 The $\bar{x}$ and s control charts for piston-ring data with variable sample size, Example 5-4.

s^2 Control Chart

Sometimes it is desired to use s^2 chart over s chart.
The parameters for s^2 chart are:

$$\text{UCL} = \frac{\bar{s}^2}{n-1}\chi^2_{\alpha/2,n-1}$$
$$\text{Center line} = \bar{s}^2$$
$$\text{LCL} = \frac{\bar{s}^2}{n-1}\chi^2_{1-(\alpha/2),n-1} \quad (5\text{-}32)$$

Shewhart Control Chart for Individual Measurements

What if there is only one observation for each sample.

1. Automated inspection and measurement technology is used, and every unit manufactured is analyzed so there is no basis for rational subgrouping.
2. The production rate is very slow, and it is inconvenient to allow sample sizes of $n > 1$ to accumulate before analysis. The long interval between observations will cause problems with rational subgrouping.
3. Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
4. Multiple measurements are taken on the same unit of product, such as measuring oxide thickness at several different locations on a wafer in semiconductor manufacturing.
5. In process plants, such as papermaking, measurements on some parameter such as coating thickness *across* the roll will differ very little and produce a standard deviation that is much too small if the objective is to control coating thickness *along* the roll.

Use the moving range between two successive samples for range.

$$\mathrm{MR}_i = |x_i - x_{i-1}|$$

Example 5-5

Table 5-6 Viscosity of Aircraft Primer Paint

Batch Number	Viscosity x	Moving Range MR
1	34.05	
2	34.40	0.35
3	33.59	0.81
4	35.96	2.37
5	34.70	1.26
6	33.51	1.19
7	33.79	0.28
8	34.04	0.25
9	34.52	0.48
10	33.75	0.77
11	33.27	0.48
12	33.71	0.44
13	34.03	0.32
14	34.58	0.55
15	34.02	0.56
16	33.97	0.05
17	34.05	0.08
18	34.04	0.01
19	33.73	0.31
20	34.05	0.32
	$\bar{x} = 34.088$	$overline{MR} = 0.5726$

Use the d_2, D_3 and D_4 values from $n = 2$ row for individual measurements. Then:

$$\text{UCL} = \bar{x} + 3\frac{\overline{\text{MR}}}{d_2} = 34.088 + 3\left(\frac{0.5726}{1.128}\right) = 35.61$$

$$\text{Center line} = \bar{x} = 34.088$$

$$\text{LCL} = \bar{x} - 3\frac{\overline{\text{MR}}}{d_2} = 34.088 - 3\left(\frac{0.5726}{1.128}\right) = 32.57$$

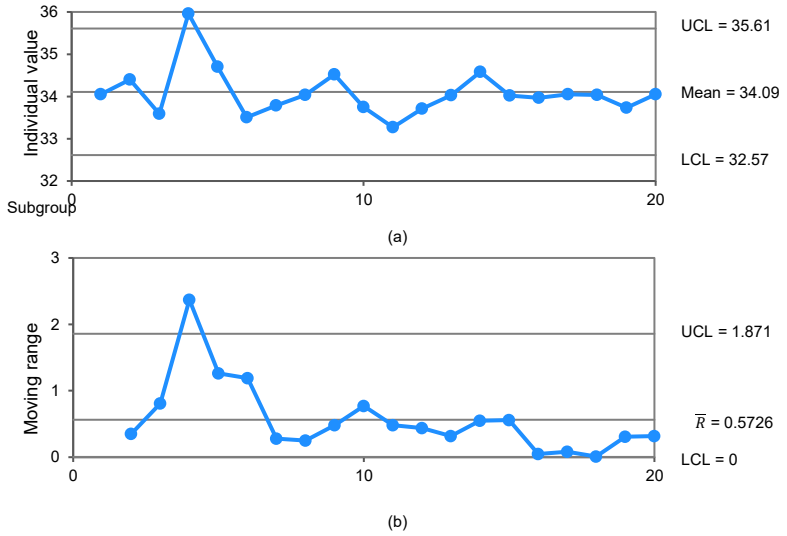


Figure 5-19 Control charts for (*a*) the moving range and for (*b*) individual observations on viscosity.

Phase II Operation and Interpretation of Charts

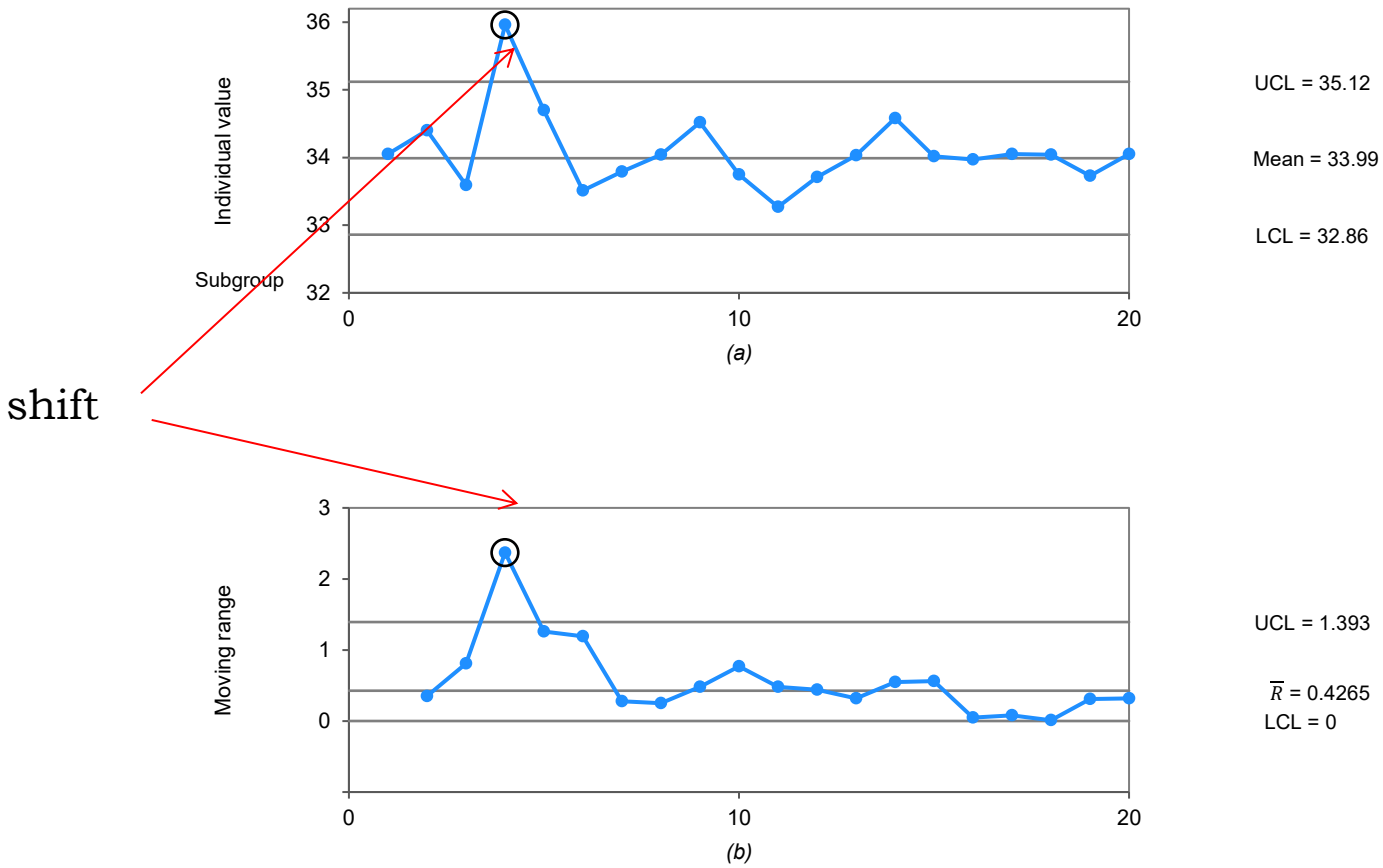


Figure 5-20 Revised individuals and moving-range control charts for viscosity.

Table 5-7 Viscosity for Aircraft Primer Paint, Batches 16-30

Batch Number	Viscosity x	Moving Range MR
21[a]	34.39	0.34
22	33.75	0.64
23	33.40	0.35
24	33.27	0.13
25	34.65	1.38
26	34.80	0.15
27	34.55	0.25
28	35.40	0.85
29	34.75	0.65
30	34.50	0.25
31	34.70	0.20
32	34.29	0.41
33	34.61	0.32
34	34.49	0.12
35	35.03	0.54

shift

[a]The moving range for batch 21 was computed as the difference in viscosity readings between batches 21 and 20; that is, $MR_{21} = |(x_{21} - x_{20})| = |(34.39 - 34.05)| = 0.34$.

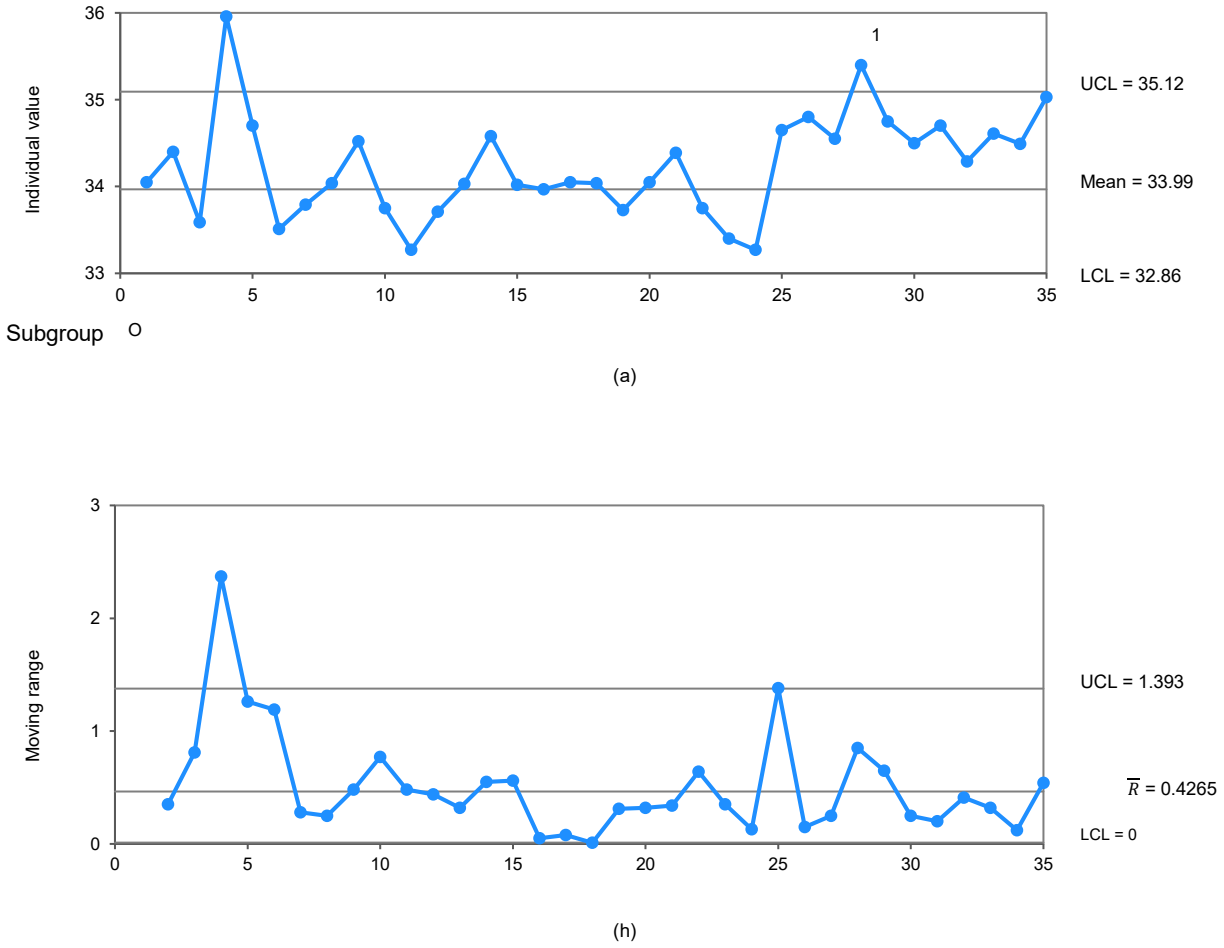


Figure 5-21 Continuation of the control chart for individuals and the moving range using the additional data in Table 5-7.

Average Run Lengths

- ARL_0 of combined individuals and moving-range chart with conventional 3-sigma limits is generally much less than Shewhart control chart.
- Ability of individuals chart to detect small shifts is very poor.

Size of Shift	β	ARL_1
1σ	0.9772	43.96
2σ	0.8413	6.30
3σ	0.5000	2.00

Normality

- In-control ARL is dramatically affected by nonnormal data.
- One approach for nonnormal data is to determine control limits for individuals control chart based on percentiles of correct underlying distribution.

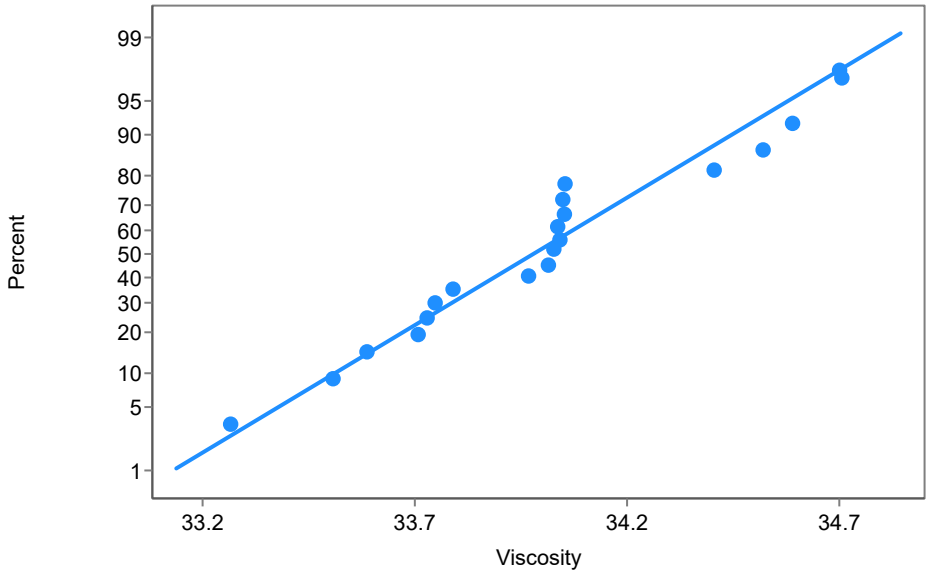


Figure 5-22 Normal probability plot of the primer paint viscosity data from Table 5-6, Example 5-5.

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Example 5-6

Table 5-8 Resistivity Data for Example 5-6

Sample, i	Resistivity (x_i)	Ln (x_i)	MR
1	216	5.37528	
2	290	5.66988	0.29460
3	236	5.46383	0.20605
4	228	5.42935	0.03448
5	244	5.49717	0.06782
6	210	5.34711	0.15006
7	139	4.93447	0.41264
8	310	5.73657	0.80210
9	240	5.48064	0.25593
10	211	5.35186	0.12878
11	175	5.16479	0.18707
12	447	6.10256	0.93777
13	307	5.72685	0.37571
14	242	5.48894	0.23791
15	168	5.12396	0.36498
16	360	5.88610	0.76214
17	226	5.42053	0.46557
18	253	5.53339	0.11286
19	380	5.94017	0.40678
20	131	4.87520	1.06497
21	173	5.15329	0.27809
22	224	5.41165	0.25836
23	195	5.27300	0.13865
24	199	5.29330	0.02030
25	226	5.42053	0.12723
		$\overline{ln(x_i)}$ = 5.44402	$overline{MR}$ = 0.33712

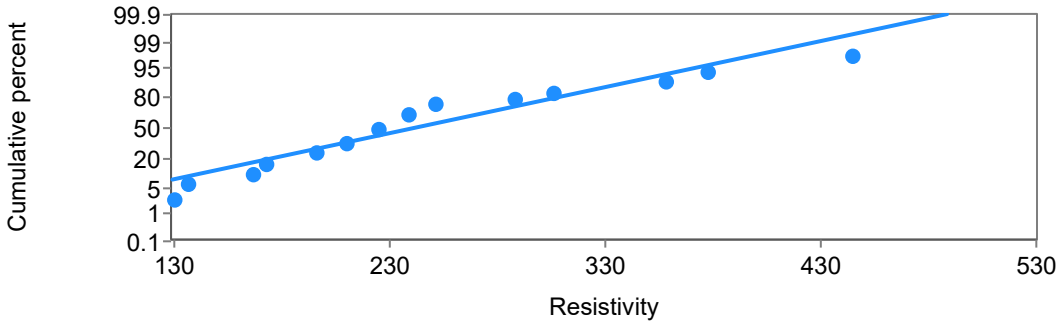


Figure 5-23 Normal probability plot of resistivity.

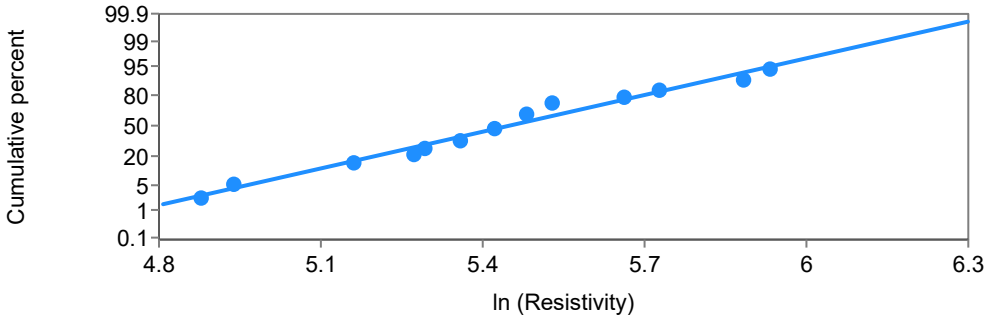


Figure 5-24 Normal probability plot of ln (resistivity).

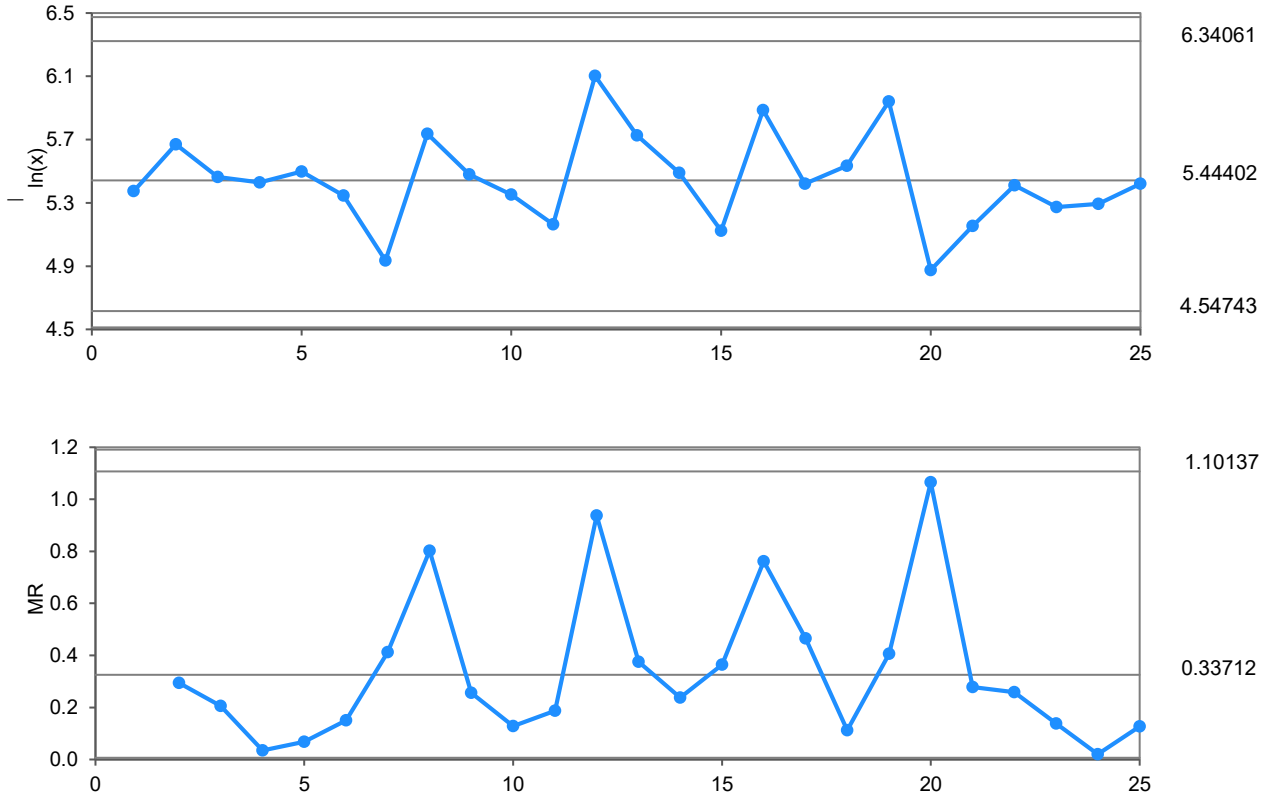


Figure 5-25 Individuals and moving range control charts on ln(resistivity), Example 5-6.

Skip Section on More about Estimating σ (pages 239 – 242).

Control Charts for Attributes

6 Control Charts for Attributes

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CHAPTER OUTLINE

6-1 INTRODUCTION

6-2 THE CONTROL CHART FOR FRACTION NONCONFORMING

- 6-2.1 Development and Operation of the Control Chart
- 6-2.2 Variable Sample Size
- 6-2.3 Nonmanufacturing Applications
- 6-2.4 The Operating-Characteristic Function and Average Run Length Calculations

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

- 6-3.1 Procedures with Constant Sample Size
- 6-3.2 Procedures with Variable Sample Size
- 6-3.3 Demerit Systems
- 6-3.4 The Operating-Characteristic Function
- 6-3.5 Dealing with Low Defect Levels
- 6-3.6 Nonmanufacturing Applications of the Control Chart

6-4 CHOICE BETWEEN ATTRIBUTES AND VARIABLES CONTROL CHARTS

6-5 GUIDELINES FOR IMPLEMENTING CONTROL CHARTS

Supplemental Material for Chapter 6

- S6-1 Probability Limits on Control Charts

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LEARNING OBJECTIVES

1. Understand the statistical basis of attributes control charts
2. Know how to design attributes control charts
3. Know how to set up and use the p chart for fraction nonconforming
4. Know how to set up and use the np control chart for the number of nonconforming items
5. Know how to set up and use the c control chart for defects
6. Know how to set up and use the u control chart for defects per unit
7. Use attributes control charts with variable sample size
8. Understand the advantages and disadvantages of attributes versus variables control charts
9. Understand the rational subgroup concept for attributes control charts
10. Determine the average run length for attributes control charts

6-2 THE CONTROL CHART FOR FRACTION NONCONFORMING

The statistical principles underlying the control chart for fraction nonconforming are based on the binomial distribution. Suppose the production process is operating in a stable manner, such that the probability that any unit will not conform to specifications is p, and that successive units produced are independent. Then each unit produced is a realization of a Bernoulli random variable with parameter p. If a random sample of n units of product is selected, and if D is the number of units of product that are nonconforming, then D has a binomial distribution with parameters n and p; that is,

$$P\{D = x\} = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0,1,\ldots,n \tag{6-1}$$

From Section 2-2.2 we know that the mean and variance of the random variable D are np and $np(1-p)$, respectively.

6-2 THE CONTROL CHART FOR FRACTION NONCONFORMING

Recall: A quality characteristic follows a binomial distribution if:

1. All trials are independent.
2. Each outcome is either a “success” or “failure”.
3. The probability of success on any trial is given as p. The probability of a failure is 1-p.
4. The probability of a success is constant.

The **sample fraction nonconforming** is defined as the ratio of the number of nonconforming units in the sample D to the sample size n; that is,

$$\hat{p} = \frac{D}{n} \tag{6-2}$$

As noted in Section 2-2.2, the distribution of the random variable $\hat{p}$ can be obtained from the binomial. Furthermore, the mean and variance of $\hat{p}$ are

$$\mu = p \tag{6-3}$$

and

$$\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n} \tag{6-4}$$

respectively.

If w is a statistic that measures a quality characteristic, and if the mean of w is μ_w and the variance of w is σ_w^2, then the general model for the Shewhart control chart is as follows:

$$\begin{aligned} \text{UCL} &= \mu_w + L\sigma_w \\ \text{Center line} &= \mu_w \\ \text{LCL} &= \mu_w - L\sigma_w \end{aligned} \tag{6-5}$$

Fraction Nonconforming Control Chart: Standard Given

$$\begin{aligned} \text{UCL} &= p + 3\sqrt{\frac{p(1-p)}{n}} \\ \text{Center line} &= p \\ \text{LCL} &= p - 3\sqrt{\frac{p(1-p)}{n}} \end{aligned} \tag{6-6}$$

When the process fraction nonconforming p is not known, then it must be estimated from observed data. The usual procedure is to select m preliminary samples, each of size n. As a general rule, m should be 20 or 25. Then if there are D_i nonconforming units in sample i, we compute the fraction nonconforming in the ith sample as

$$\hat{p}_i = \frac{D_i}{n} \qquad i = 1,2,\ldots,m$$

and the average of these individual sample fractions nonconforming is

$$\overline{p} = \frac{\sum_{i=1}^{m} D_i}{mn} = \frac{\sum_{i=1}^{m} \hat{p}_i}{m} \tag{6-7}$$

The statistic $\overline{p}$ estimates the unknown fraction nonconforming p. The center line and control limits of the control chart for fraction nonconforming are computed as follows:

Fraction Nonconforming Control Chart: No Standard Given

$$\text{UCL} = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

$$\text{Center line} = \overline{p} \qquad (6\text{-}8)$$

$$\text{LCL} = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

EXAMPLE 6-1

Frozen orange juice concentrate is packed in 6-oz cardboard cans. These cans are formed on a machine by spinning them from cardboard stock and attaching a metal bottom panel. By inspection of a can, we may determine whether, when filled, it could possibly leak either on the side seam or around the bottom joint. Such a nonconforming can has an improper seal on either the side seam or the bottom panel. We wish to set up a control chart to improve the fraction of nonconforming cans produced by this machine.

To establish the control chart, 30 samples of $n = 50$ cans each were selected at half-hour intervals over a three-shift period in which the machine was in continuous operation. The data are shown in Table 6-1.

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EXAMPLE 6-1 (cont'd)

Table 6-1 Data for Trial Control Limits, Example 6-1, Sample Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
1	12	0.24	16	8	0.16
2	15	0.30	17	10	0.20
3	8	0.16	18	5	0.10
4	10	0.20	19	13	0.26
5	4	0.08	20	11	0.22
6	7	0.14	21	20	0.40
7	16	0.32	22	18	0.36
8	9	0.18	23	24	0.48
9	14	0.28	24	15	0.30
10	10	0.20	25	9	0.18
11	5	0.10	26	12	0.24
12	6	0.12	27	7	0.14
13	17	0.34	28	13	0.26
14	12	0.24	29	9	0.18
15	22	0.44	30	6	0.12

We construct a phase I control chart using this preliminary data to determine if the process was in control when these data were collected. Since the 30 samples contain $\sum_{i=1}^{30} D_i = 347$ nonconforming cans, we find from equation 6-7,

$$\overline{p} = \frac{\sum_{i=1}^{m} D_i}{mn} = \frac{347}{(30)(50)} = 0.2313$$

Using $\overline{p}$ as an estimate of the true process fraction nonconforming, we can now calculate the upper and lower control limits as

$$\overline{p} \pm 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.2313 \pm 3\sqrt{\frac{0.2313(0.7687)}{50}}$$
$$= 0.2313 \pm 3(0.0596)$$
$$= 0.2313 \pm 0.1789$$

$$\text{UCL} = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.2313 + 0.1789 = 0.4102$$

$$\text{LCL} = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.2313 - 0.1789 = 0.0524$$

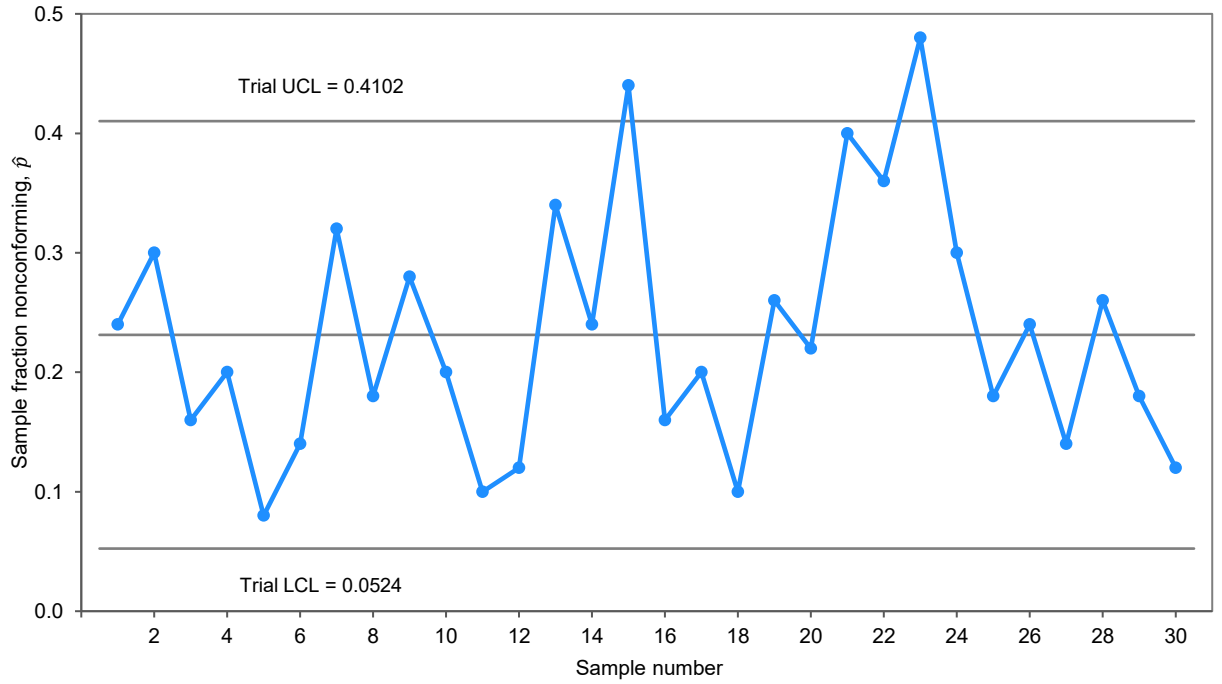


Figure 6-1 Initial phase I fraction nonconforming control chart for the data in Table 6-1.

samples 15 and 23 are eliminated, and the new center line and revised control limits are calculated as

$$\overline{p} = \frac{301}{(28)(50)} = 0.2150$$

$$\text{UCL} = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.2150 + 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.3893$$

$$\text{LCL} = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.2150 - 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.0407$$

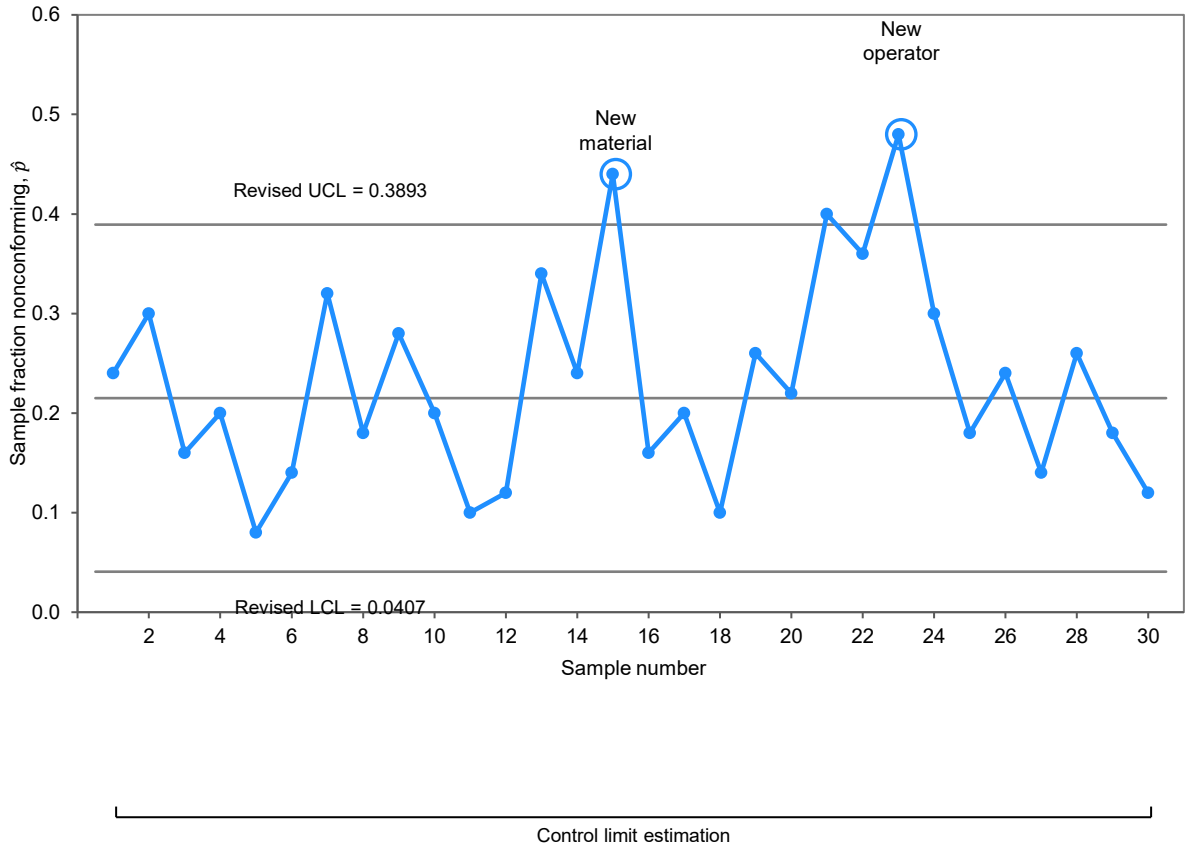


Figure 6-2 Revised control limits for the data in Table 6-1.

During the next three shifts following the machine adjustments and the introduction of the control chart, an additional 24 samples of $n = 50$ observations each are collected. These data are shown in Table 6-2, and the sample fractions nonconforming are plotted on the control chart in Fig. 6-3.

Table 6-2 Orange Juice Concentrate Can Data in Samples of Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
31	9	0.18	43	3	0.06
32	6	0.12	44	6	0.12
33	12	0.24	45	5	0.10
34	5	0.10	46	4	0.08
35	6	0.12	47	8	0.16
36	4	0.08	48	5	0.10
37	6	0.12	49	6	0.12
38	3	0.06	50	7	0.14
39	7	0.14	51	5	0.10
40	6	0.12	52	6	0.12
41	2	0.04	53	3	0.06
42	4	0.08	54	5	0.10

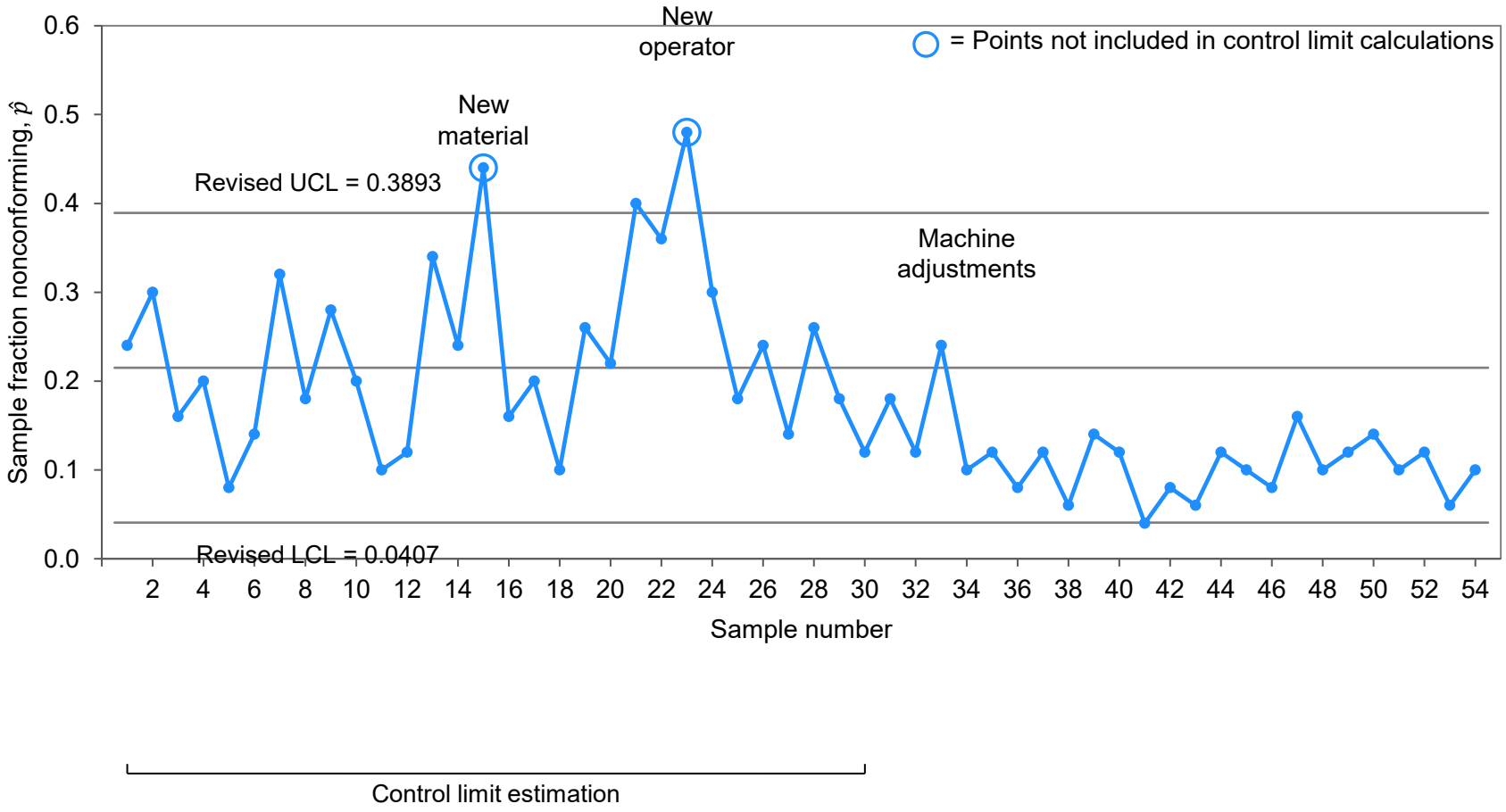


Figure 6-3 Continuation of the fraction nonconforming control chart, Example 6-1.

Based on the apparently successful process adjustments, it seems logical to revise the control limits again, using only the most recent samples (numbers 31–54). This results in the new control chart parameters:

$$\text{Center line} = \bar{p} = 0.1108$$

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 + 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = 0.2440$$

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 - 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = -0.0224 = 0$$

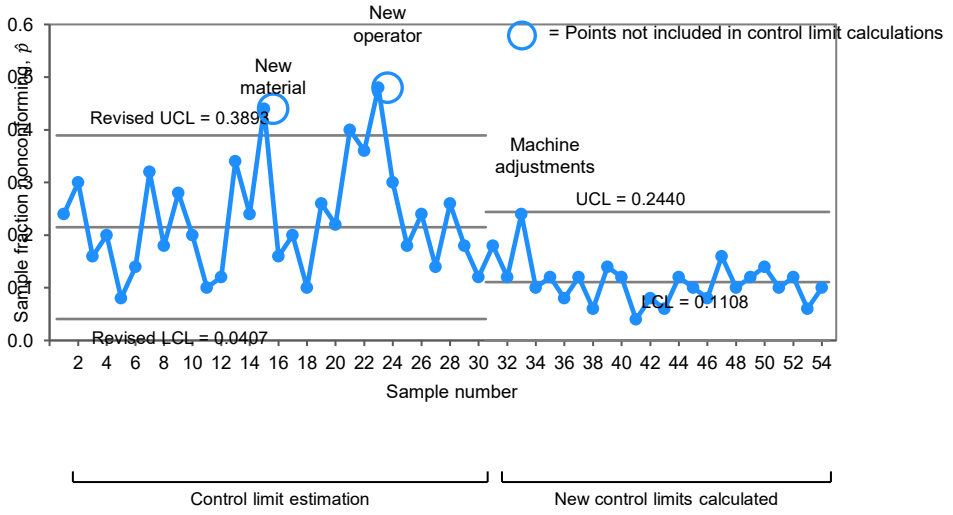


Figure 6-4 New control limits on the fraction nonconforming control chart, Example 6-1.

Table 6-3 New Data for the Fraction Nonconforming Control Chart in Fig. 6-5, $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
55	8	0.16	75	5	0.10
56	7	0.14	76	8	0.16
57	5	0.10	77	11	0.22
58	6	0.12	78	9	0.18
59	4	0.08	79	7	0.14
60	5	0.10	80	3	0.06
61	2	0.04	81	5	0.10
62	3	0.06	82	2	0.04
63	4	0.08	83	1	0.02
64	7	0.14	84	4	0.08
65	6	0.12	85	5	0.10
66	5	0.10	86	3	0.06
67	5	0.10	87	7	0.14
68	3	0.06	88	6	0.12
69	7	0.14	89	4	0.08
70	9	0.18	90	4	0.08
71	6	0.12	91	6	0.12
72	10	0.20	92	8	0.16
73	4	0.08	93	5	0.10
74	3	0.06	94	6	0.12

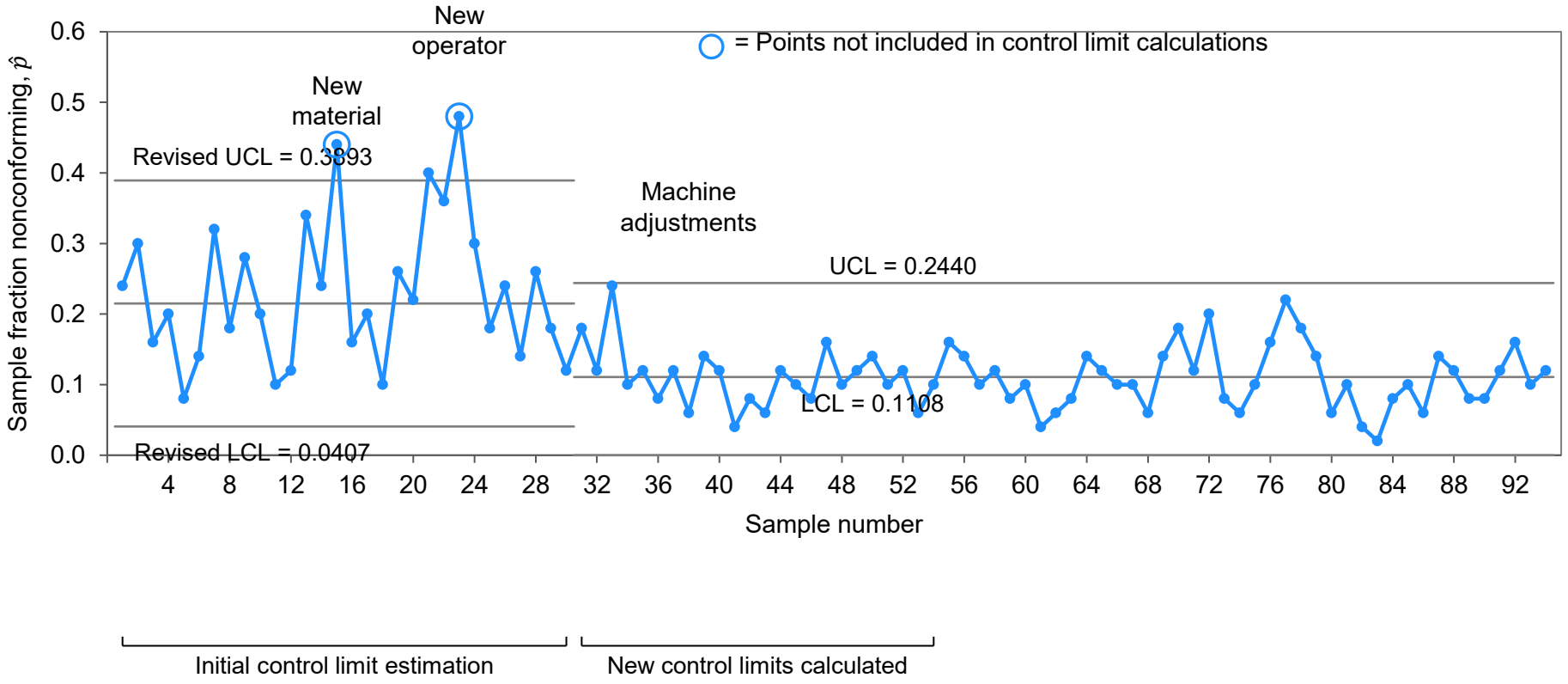


Figure 6-5 Completed fraction nonconforming control chart, Example 6-1.

Design of Fraction Nonconforming Chart

- Three parameters must be specified
  1. The sample size
  2. The frequency of sampling
  3. The width of the control limits
- Common to base chart on 100% inspection of *all* process output over time
- Rational subgroups may also play role in determining sampling frequency

Interpretation of Points on the Control Chart for Fraction Nonconforming

Example 6-1 illustrated how points that plot beyond the control limits are treated, both in establishing the control chart and during its routine operation. Care must be exercised in interpreting points that plot *below* the lower control limit. These points often do not represent a real improvement in process quality. Frequently, they are caused by errors in the inspection process resulting from inadequately trained or inexperienced inspectors or from improperly calibrated test and inspection equipment. We have also seen cases in which inspectors deliberately passed nonconforming units or reported fictitious data. The analyst must keep these warnings in mind when looking for assignable causes if points plot below the lower control limits. Not all "downward shifts" in p are attributable to improved quality.

The np Control Chart

It is also possible to base a control chart on the number nonconforming rather than the fraction nonconforming. This is often called an np control chart. The parameters of this chart are as follows.

The np Control Chart

$$\text{UCL} = np + 3\sqrt{np(1-p)}$$
$$\text{Center line} = np$$
$$\text{LCL} = np - 3\sqrt{np(1-p)} \quad (6\text{-}13)$$

6-2.2 Variable Sample Size

Variable-Width Control Limits

The first and perhaps the most simple approach is to determine control limits for each individual sample that are based on the specific sample size. That is, if the ith sample is of size n_i, then the upper and lower control limits are $\overline{p} \pm 3\sqrt{\overline{p}(1-\overline{p})/n_i}$.

Table 6-4 Data for a Control Chart for Fraction Nonconforming with Variable Sample Size

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_{\hat{p}} = \sqrt{((0.096)(0.904)/n_i)}$	Control Limits LCL	
1	100	12	0.120	0.029	0.009	0.183
2	80	8	0.100	0.033	0	0.195
3	80	6	0.075	0.033	0	0.195
4	100	9	0.090	0.029	0.009	0.183
5	110	10	0.091	0.028	0.012	0.180
6	110	12	0.109	0.028	0.012	0.180
7	100	11	0.110	0.029	0.009	0.183
8	100	16	0.160	0.029	0.009	0.183
9	90	10	0.110	0.031	0.003	0.189
10	90	6	0.067	0.031	0.003	0.189
11	110	20	0.182	0.028	0.012	0.180
12	120	15	0.125	0.027	0.015	0.177
13	120	9	0.075	0.027	0.015	0.177

Table 6-4 (cont'd) Data for a Control Chart for Fraction Nonconforming with Variable Sample Size

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_{\hat{p}} = \sqrt{((0.096)(0.904)/n_i)}$	Control Limits LCL	
14	120	8	0.067	0.027	0.015	0.177
15	110	6	0.055	0.028	0.012	0.180
16	80	8	0.100	0.033	0	0.195
17	80	10	0.125	0.033	0	0.195
18	80	7	0.088	0.033	0	0.195
19	90	5	0.056	0.031	0.003	0.189
20	100	8	0.080	0.029	0.009	0.183
21	100	5	0.050	0.029	0.009	0.183
22	100	8	0.080	0.029	0.009	0.183
23	100	10	0.100	0.029	0.009	0.183
24	90	6	0.067	0.031	0.003	0.189
25	90	9	0.100	0.031	0.003	0.189
	2450	234	2.383			

To illustrate this approach, consider the data in Table 6-4. For the 25 samples, we calculate

$$\overline{p} = \frac{\sum_{i=1}^{25} D_i}{\sum_{i=1}^{25} n_i} = \frac{234}{2450} = 0.096$$

Consequently, the center line is at 0.096, and the control limits are

$$\text{UCL} = \overline{p} + 3\hat{\sigma}_{\hat{p}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

and

$$\text{LCL} = \overline{p} - 3\hat{\sigma}_{\hat{p}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

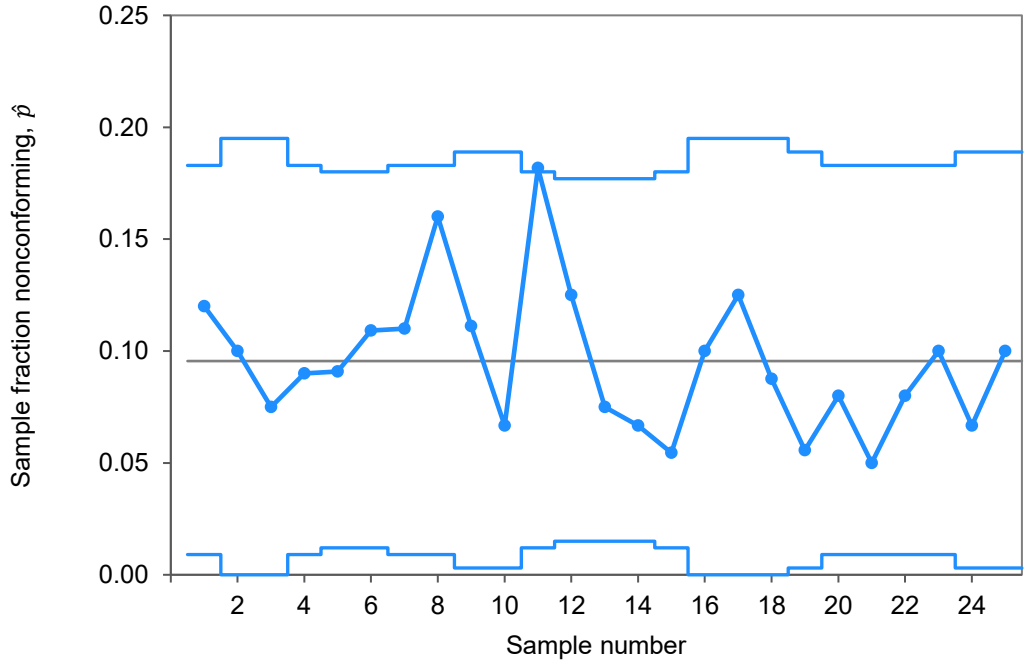


Figure 6-6 Control chart for fraction nonconforming with variable sample size.

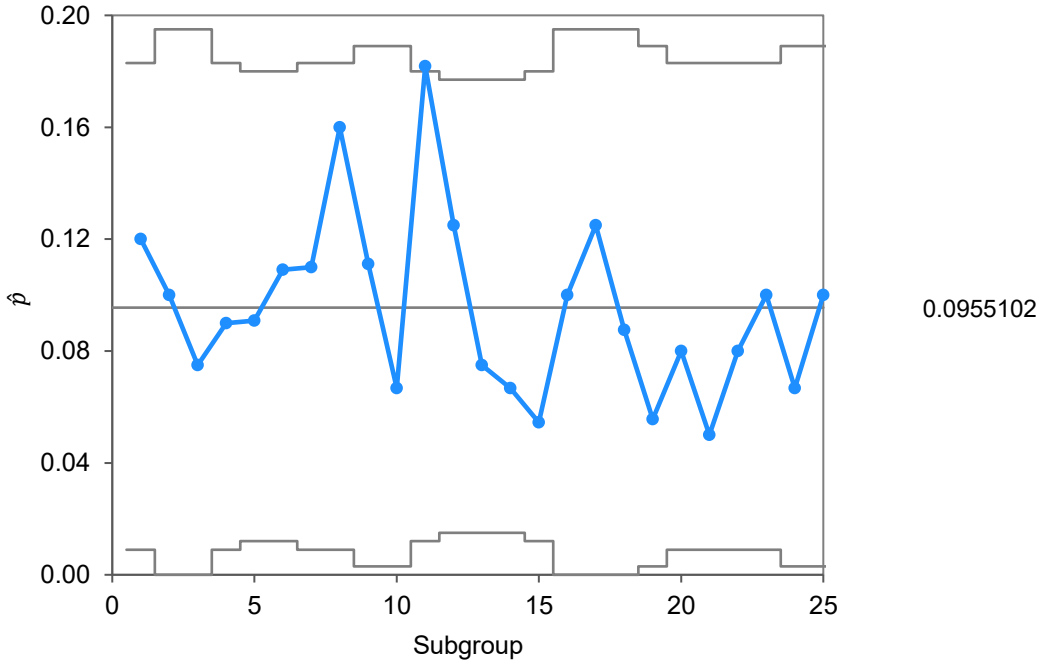


Figure 6-7 Control chart for fraction nonconforming with variable sample size using Minitab.

6-2.2 Variable Sample Size

Control Limits Based on an Average Sample Size

The second approach is to base the control chart on an *average* sample size, resulting in an **approximate set of control limits.** This assumes that future sample sizes will not differ greatly from those previously observed.

For the data in Table 6-4, we find that the average sample size is

$$\bar{n} = \frac{\sum_{i=1}^{25} n_i}{25} = \frac{2450}{25} = 98$$

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.185$$

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.007$$

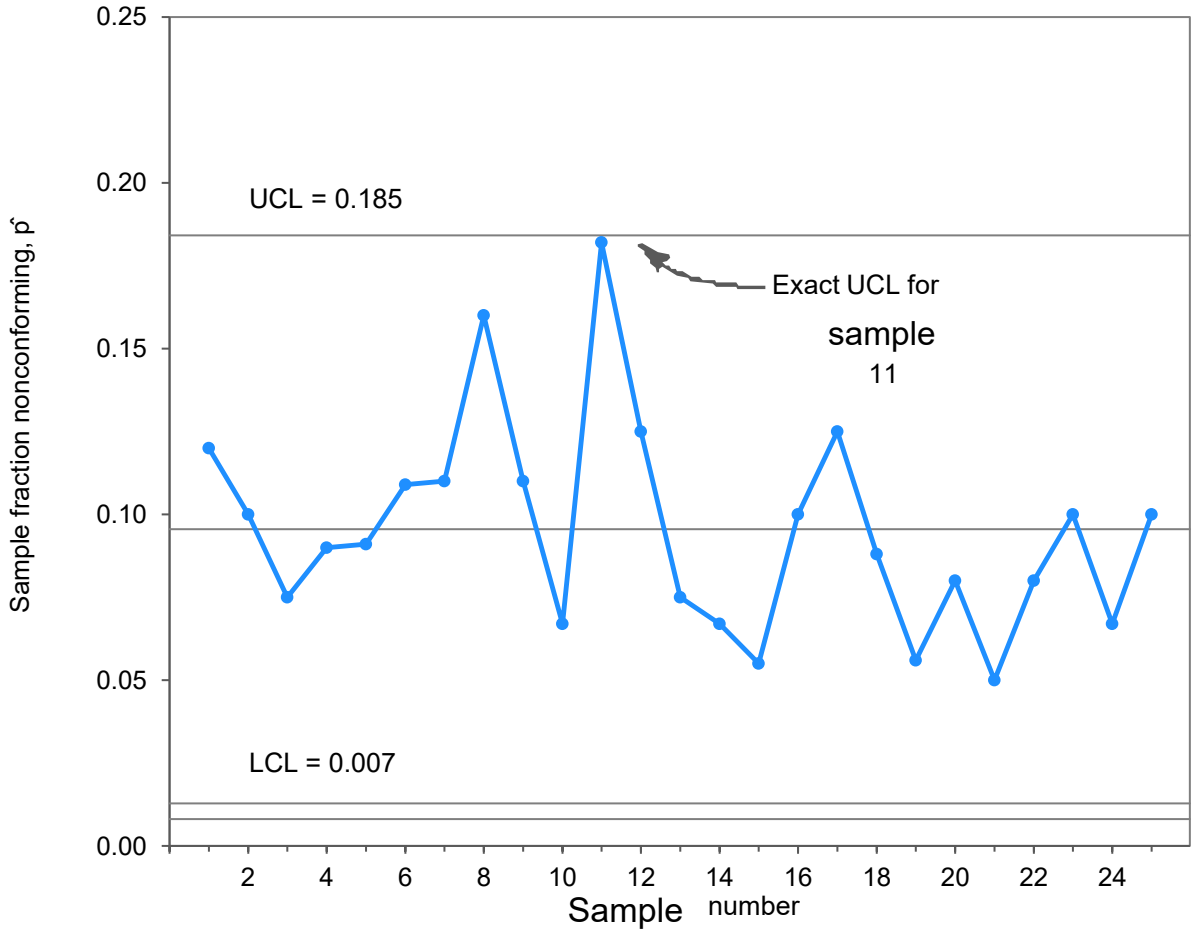


Figure 6-8 Control chart for fraction nonconforming based on average sample size.

6-2.2 Variable Sample Size

The Standardized Control Chart

The third approach to dealing with variable sample size is to use a "standardized" control chart, where the points are plotted in standard deviation units. Such a control chart has the center line at zero, and upper and lower control limits of +3 and −3, respectively. The variable plotted on the chart is

$$Z_i = \frac{\hat{p}_i - p}{\sqrt{\frac{p(1-p)}{n_i}}} \tag{6-14}$$

where p (or $\overline{p}$ if no standard is given) is the process fraction nonconforming in the in-control state. The standardized control chart for the data in Table 6-4 is shown in Fig. 6-9.

Table 6-5 Calculations for the Standardized Control Chart in Fig. 6-9, $\bar{p} = 0.096$

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_{\hat{p}} = \sqrt{((0.096)(0.904)/n_i)}$	$Z_i = (\hat{p}_i - \bar{p})/(\sqrt{((0.096)(0.904)/n_i)})$
1	100	12	0.120	0.029	0.83
2	80	8	0.100	0.033	0.12
3	80	6	0.075	0.033	−0.64
4	100	9	0.090	0.029	−0.21
5	110	10	0.091	0.028	−0.18
6	110	12	0.109	0.028	0.46
7	100	11	0.110	0.029	0.48
8	100	16	0.160	0.029	2.21
9	90	10	0.110	0.031	0.45
10	90	6	0.067	0.031	−0.94
11	110	20	0.182	0.028	3.07
12	120	15	0.125	0.027	1.07
13	120	9	0.075	0.027	−0.78

Table 6-5 (cont'd) Calculations for the Standardized Control Chart in Fig. 6-9, $\bar{p} = 0.096$

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_{\hat{p}} = \sqrt{((0.096)(0.904)/n_i)}$	$Z_i = (\hat{p}_i - \bar{p})/(\sqrt{((0.096)(0.904)/n_i)})$
14	120	8	0.067	0.027	−1.07
15	110	6	0.055	0.028	−1.46
16	80	8	0.100	0.033	0.12
17	80	10	0.125	0.033	0.88
18	80	7	0.088	0.033	−0.24
19	90	5	0.056	0.031	−1.29
20	100	8	0.080	0.029	−0.55
21	100	5	0.050	0.029	−1.59
22	100	8	0.080	0.029	−0.55
23	100	10	0.100	0.029	0.14
24	90	6	0.067	0.031	−0.94
25	90	9	0.100	0.031	0.13

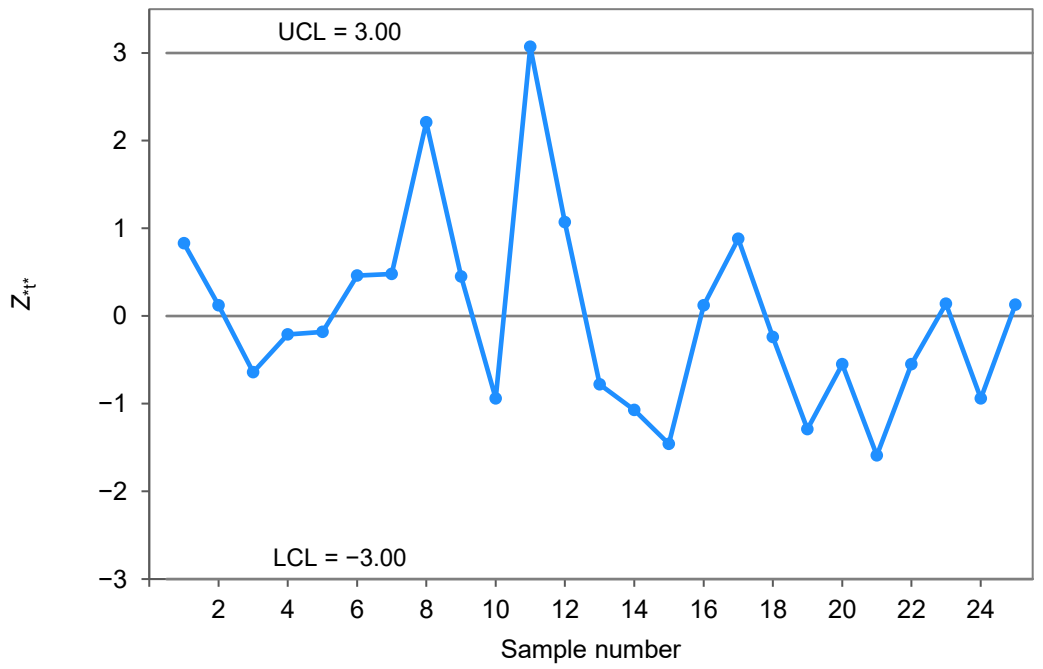


Figure 6-9 Standardized control chart for fraction nonconforming.

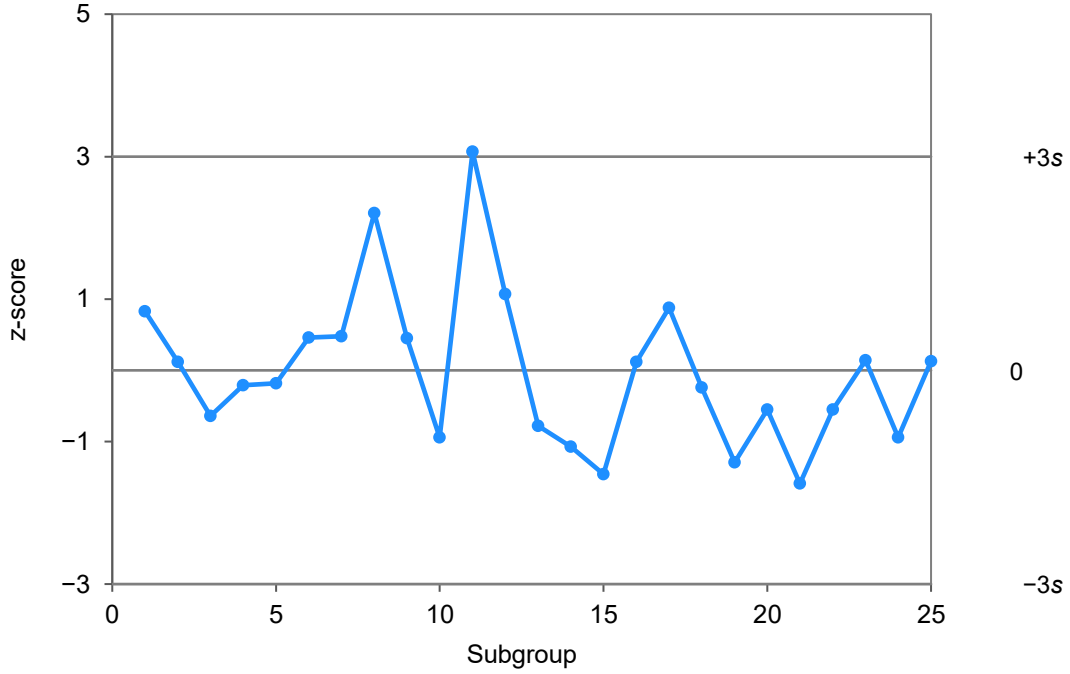


Figure 6-10 Standardized control chart from Minitab for fraction nonconforming, Table 6-4.

6-2.4 The Operating-Characteristic Function and Average Run Length Calculations

The probability of type II error for the fraction nonconforming control chart may be computed from

$$\beta = P\{\hat{p} < \text{UCL} \mid p\} - P\{\hat{p} \leq \text{LCL} \mid p\}$$
$$= P\{D < n\text{UCL} \mid p\} - P\{D \leq n\text{LCL} \mid p\} \quad (6\text{-}15)$$

Table 6-6 illustrates the calculations required to generate the OC curve for a control chart for fraction nonconforming with parameters n = 50, LCL = 0.0303, and UCL = 0.3697. Using these parameters, equation 6-15 becomes

$$\beta = P\{D < (50)(0.3697) \mid p\} - P\{D \leq (50)(0.0303) \mid p\}$$
$$= P\{D < 18.49 \mid p\} - P\{D \leq 1.52 \mid p\}$$

However, since D must be an integer, we find that

$$\beta = P\{D \leq 18 \mid p\} - P\{D \leq 1 \mid p\}$$

The OC curve is plotted in Fig. 6-11.

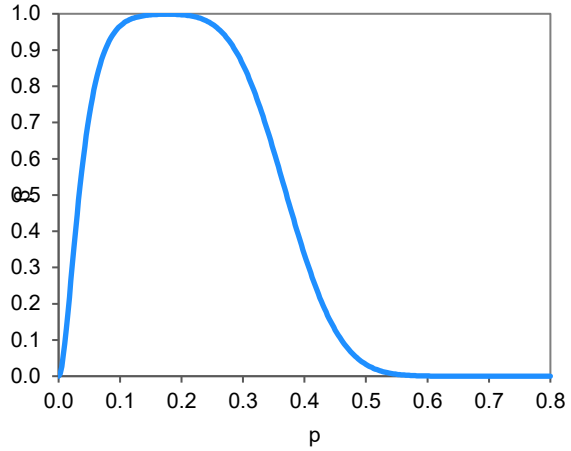


Figure 6-11 Operating-characteristic curve for the fraction nonconforming control chart with $\overline{p}$ = 0.20, LCL = 0.0303, and UCL = 0.3697.

Figure 6-11 Operating-characteristic curve for the fraction nonconforming control chart with $\overline{p}$ = 0.20, LCL = 0.0303, and UCL = 0.3697.

Table 6-6 Calculations[a] for Constructing the OC Curve for a Control Chart for Fraction Nonconforming with n = 50, LCL = 0.0303, and UCL = 0.3697

p	$P\{D \le 18 \mid p\}$	$P\{D \le 1 \mid p\}$	$\beta = P\{D \le 18 \mid p\} - P\{D \le 1 \mid p\}$
0.01	1.0000	0.9106	0.0894
0.03	1.0000	0.5553	0.4447
0.05	1.0000	0.2794	0.7206
0.10	1.0000	0.0338	0.9662
0.15	0.9999	0.0029	0.9970
0.20	0.9975	0.0002	0.9973
0.25	0.9713	0.0000	0.9713
0.30	0.8594	0.0000	0.8594
0.35	0.6216	0.0000	0.6216
0.40	0.3356	0.0000	0.3356
0.45	0.1273	0.0000	0.1273
0.50	0.0325	0.0000	0.0325
0.55	0.0053	0.0000	0.0053

[a]The probabilities in this table were found by evaluating the cumulative binomial distribution. For small p (p < 0.1, say) the Poisson approximation could be used, and for larger values of p the normal approximation could be used.

We may also calculate average run lengths (ARLs) for the fraction nonconforming control chart. Recall from Chapter 4 that for uncorrelated process data the ARL for any Shewhart control chart can be written as

$$\text{ARL} = \frac{1}{P(\text{sample point plots out of control})}$$

Thus, if the process is in control, ARL_0 is

$$\text{ARL}_0 = \frac{1}{\alpha}$$

and if it is out of control, then

$$\text{ARL}_1 = \frac{1}{1-\beta}$$

To illustrate, consider the control chart for fraction nonconforming used in the OC curve calculations in Table 6-6. This chart has parameters n = 50, UCL = 0.3697, LCL = 0.0303, and the center line is $\overline{p}$ = 0.20. From Table 6-6 (or the OC curve in Fig. 6-11) we find that if the process is in control with $p = \overline{p}$, the probability of a point plotting in control is 0.9973. Thus, in this case $\alpha = 1 - \beta = 0.0027$, and the value of ARL_0 is

$$\text{ARL}_0 = \frac{1}{\alpha} = \frac{1}{0.0027} \simeq 370$$

Therefore, if the process is really in control, we will experience a "false-alarm" out-of-control signal about every 370 samples. (This will be approximately true, in general, for any Shewhart control chart with three-sigma limits.) This in-control ARL_0 is generally considered to be satisfactorily large. Now suppose that the process shifts out of control to p = 0.3. Table 6-6 indicates that if p = 0.3, then β = 0.8594. Therefore, the value of ARL_1 is

$$\text{ARL}_1 = \frac{1}{1 - \beta} = \frac{1}{1 - 0.8594} \simeq 7$$

and it will take about seven samples, on the average, to detect this shift with a point outside of the control limits.

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

6-3.1 Procedures with Constant Sample Size

Consider the occurrence of nonconformities in an inspection unit of product. In most cases, the inspection unit will be a single unit of product, although this is not necessarily always so. The inspection unit is simply an entity for which it is convenient to keep records. It could be a group of 5 units of product, 10 units of product, and so on. Suppose that defects or nonconformities occur in this inspection unit according to the Poisson distribution; that is,

$$p(x) = \frac{e^{-c}c^{x}}{x!} \qquad x = 0,1,2, \ldots$$

where x is the number of nonconformities and $c > 0$ is the parameter of the Poisson distribution.

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

- **Poisson distribution** is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time and/or space if these events occur with a known average rate and independently of the time since the last event. (The Poisson distribution can also be used for the number of events in other specified intervals such as distance, area or volume.)

Therefore, a control chart for nonconformities with three-sigma limits[2] would be defined as follows,

Control Chart for Nonconformities: Standard Given

$$\text{UCL} = c + 3\sqrt{c}$$
$$\text{Center line} = c$$
$$\text{LCL} = c - 3\sqrt{c} \qquad (6\text{-}16)$$

[2]The α-risk for three-sigma limits is not equally allocated above the UCL and below the LCL, because the Poisson distribution is asymmetric. Some authors recommend the use of probability limits for this chart, particularly when c is small.

assuming that a standard value for c is available. Should these calculations yield a negative value for the LCL, set LCL = 0.

If no standard is given, then c may be estimated as the observed average number of nonconformities in a preliminary sample of inspection units—say, $\bar{c}$. In this case, the control chart has parameters defined as follows.

Control Chart for Nonconformities: No Standard Given

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}}$$
$$\text{Center line} = \bar{c} \qquad (6\text{-}17)$$
$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}}$$

When no standard is given, the control limits in equation 6-17 should be regarded as *trial* control limits, and the preliminary samples examined for lack of control in the usual phase I analysis. The control chart for nonconformities is also sometimes called the c chart.

EXAMPLE 6-3

Table 6-7 presents the number of nonconformities observed in 26 successive samples of 100 printed circuit boards. Note that, for reasons of convenience, the inspection unit is defined as 100 boards.

Table 6-7 Data on the Number of Nonconformities in Samples of 100 Printed Circuit Boards

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	21	14	19
2	24	15	10
3	16	16	17
4	12	17	13
5	15	18	22
6	5	19	18
7	28	20	39
8	20	21	30
9	31	22	24
10	25	23	16
11	20	24	19
12	24	25	17
13	16	26	15

Since the 26 samples contain 516 total nonconformities, we estimate c by

$$\bar{c} = \frac{516}{26} = 19.85$$

Therefore, the trial control limits are given by

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 19.85 + 3\sqrt{19.85} = 33.22$$
$$\text{Center line} = \bar{c} = 19.85$$
$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} = 19.85 - 3\sqrt{19.85} = 6.48$$

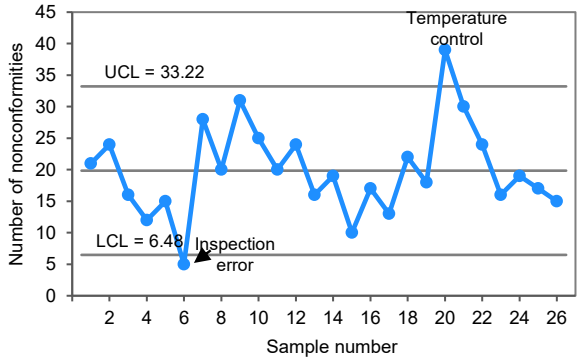


Figure 6-12 Control chart for nonconformities for Example 6-3.

The control chart is shown in Fig. 6-12. The number of observed nonconformities from the preliminary samples is plotted on this chart. Two points plot outside the control limits, samples 6 and 20. Investigation of sample 6 revealed that a new inspector had examined the boards in this sample and that he did not recognize several of the types of nonconformities that could have been present. Furthermore, the unusually large number of nonconformities in sample 20 resulted from a temperature control problem in the wave soldering machine, which was subsequently repaired. Therefore, it seems reasonable to exclude these two samples and revise the trial control limits. The estimate of c is now computed as

$$\overline{c} = \frac{472}{24} = 19.67$$

and the revised control limits are

$$\text{UCL} = \overline{c} + 3\sqrt{\overline{c}} = 19.67 + 3\sqrt{19.67} = 32.97$$
$$\text{Center line} = \overline{c} = 19.67$$
$$\text{LCL} = \overline{c} - 3\sqrt{\overline{c}} = 19.67 - 3\sqrt{19.67} = 6.36$$

These become the standard values against which production in the next period can be compared.

Twenty new samples, each consisting of one inspection unit (i.e., 100 boards), are subsequently collected. The number of nonconformities in each sample is noted and recorded in Table 6-8. These points are plotted on the control chart in Fig. 6-13. No lack of control is indicated; however, the number of nonconformities per board is still unacceptably high. Management action is necessary to improve the process.

Table 6-8 Additional Data for the Control Chart for Nonconformities, Example 6-3

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
27	16	37	18
28	18	38	21
29	12	39	16
30	15	40	22
31	24	41	19
32	21	42	12
33	28	43	14
34	20	44	9
35	25	45	16
36	19	46	21

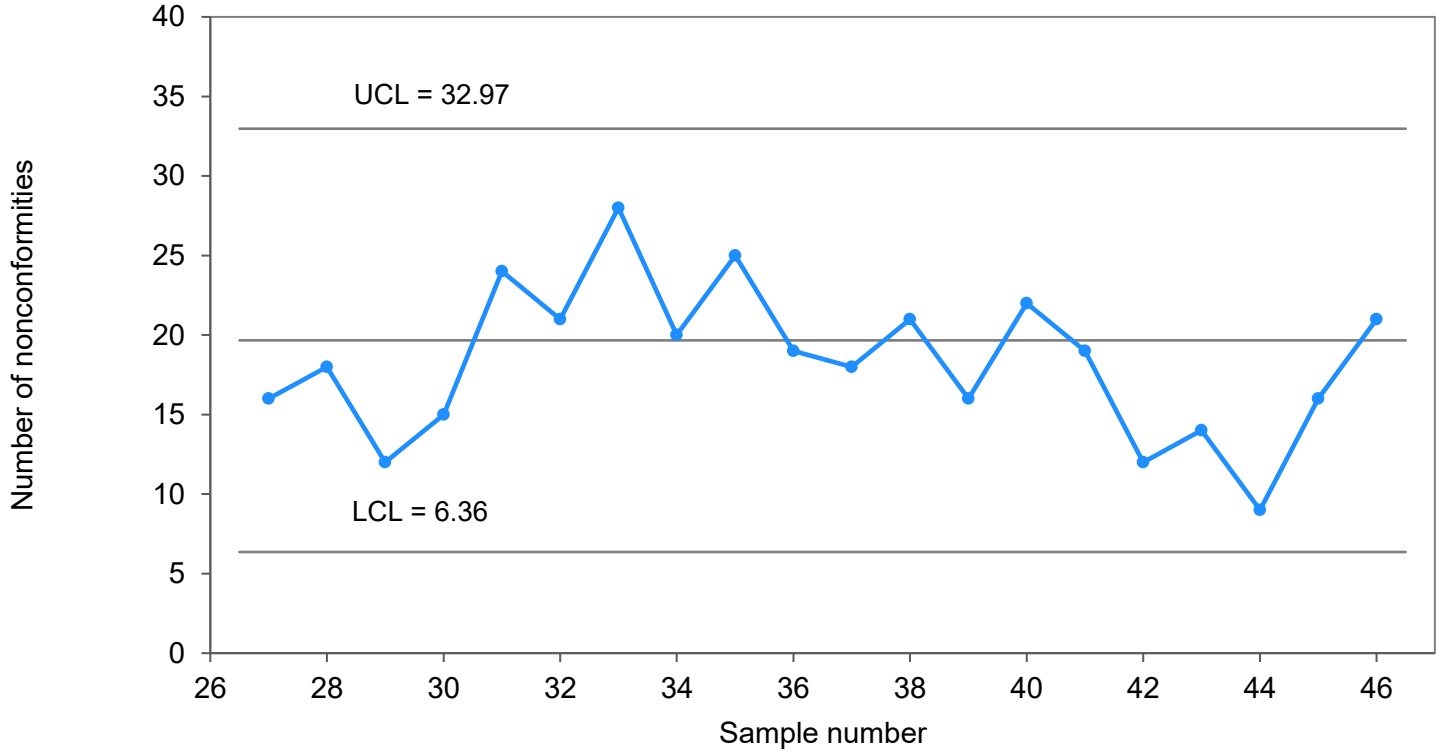


Figure 6-13 Continuation of the control chart for nonconformities, Example 6-3.

Further Analysis of Nonconformities

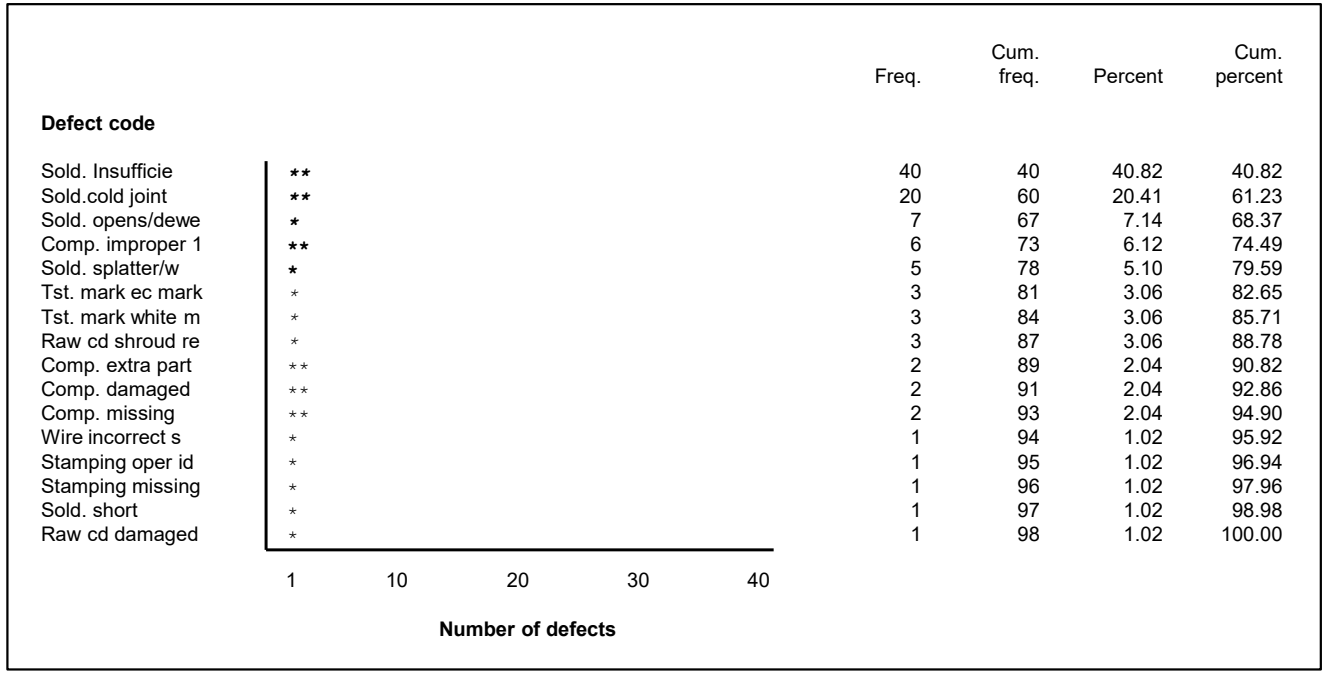


Figure 6-14 Pareto analysis of nonconformities for the printed circuit board process.

- Refer to Table 6-9 for occurrence of defect type by type of printed circuit board (part number)

Another useful technique for further analysis of nonconformities is the **cause-and-effect diagram** discussed in Chapter 4. The cause-and-effect diagram is used to illustrate the various sources of nonconformities in products and their interrelationships. It is useful in focusing the attention of operators, manufacturing engineers, and managers on quality problems. Developing a good cause-and-effect diagram usually advances the level of technological understanding of the process.

A cause-and-effect diagram for the printed circuit board assembly process is shown in Fig. 6-15. Since most of the defects in this example were solder related, the cause-and-effect diagram could be used here to help choose the variables for a designed experiment to optimize the wave soldering process.

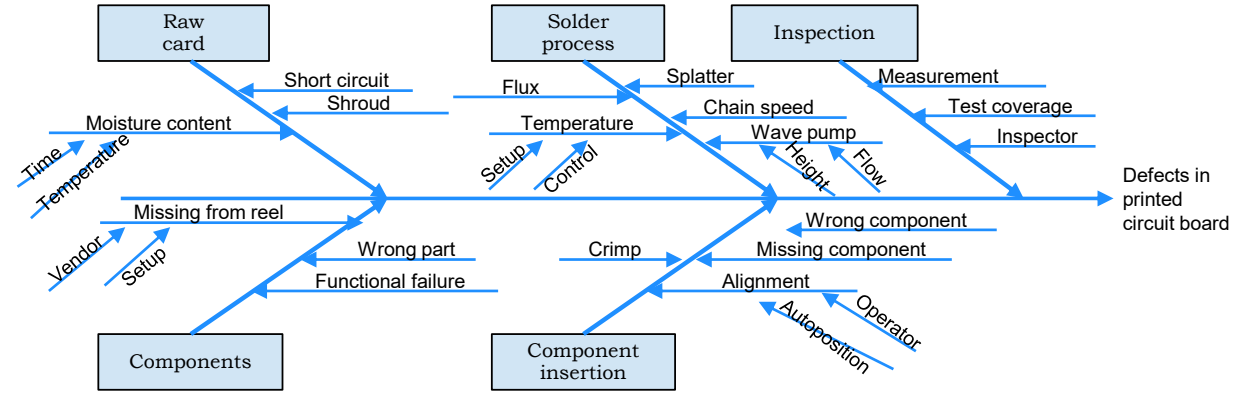


Figure 6-15 Cause-and-effect diagram.

Choice of Sample Size: The u Chart

The second approach involves setting up a control chart based on the average number of nonconformities per inspection unit. If we find x *total* nonconformities in a sample of n inspection units, then the *average* number of nonconformities per inspection unit is

$$u = \frac{x}{n} \tag{6-18}$$

Control Chart for Average Number of Nonconformities per Unit

$$\text{UCL} = \overline{u} + 3\sqrt{\frac{\overline{u}}{n}}$$
$$\text{Center line} = \overline{u} \tag{6-19}$$
$$\text{LCL} = \overline{u} - 3\sqrt{\frac{\overline{u}}{n}}$$

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EXAMPLE 6-4

A supply chain engineering group monitors shipments of materials through the company distribution network. Errors on either the delivered material or the accompanying documentation are tracked on a weekly basis. Fifty randomly selected shipments are examined and the errors recorded. Data for 20 weeks are shown in Table 6-10.

Table 6-10 Data on Number of Shipping Errors in a Supply Chain Network

Sample Number (week), i	Sample Size, n	Total Number of Errors (Nonconformities), x_i	Average Number of Errors (Nonconformities) per Unit, $u_i = x_i/n$
1	50	2	0.04
2	50	3	0.06
3	50	8	0.16
4	50	1	0.02
5	50	1	0.02
6	50	4	0.08
7	50	1	0.02
8	50	4	0.08
9	50	5	0.10
10	50	1	0.02
11	50	8	0.16
12	50	2	0.04
13	50	4	0.08
14	50	3	0.06
15	50	4	0.08
16	50	1	0.02
17	50	8	0.16
18	50	3	0.06
19	50	7	0.14
20	50	4	0.08
		74	1.48

A u control chart can be established to monitor this process. From these data, we estimate the number of errors (nonconformities) per unit (shipment) to be:

$$\overline{u} = \frac{\sum_{i=1}^{20} u_i}{20} = \frac{1.48}{20} = 0.0740$$

Therefore, the parameters of the control chart are

$$\text{UCL} = \overline{u} + 3\sqrt{\frac{\overline{u}}{n}} = 0.0740 + 3\sqrt{\frac{0.0740}{50}} = 0.1894$$

$$\text{Center line} = \overline{u} = 1.93$$

$$\text{LCL} = \overline{u} - 3\sqrt{\frac{\overline{u}}{n}} = 0.0740 - 3\sqrt{\frac{0.0740}{50}} = -0.0414$$

Since the LCL < 0, we would set LCL = 0 for the u chart. The control chart is shown in Fig. 6-16.

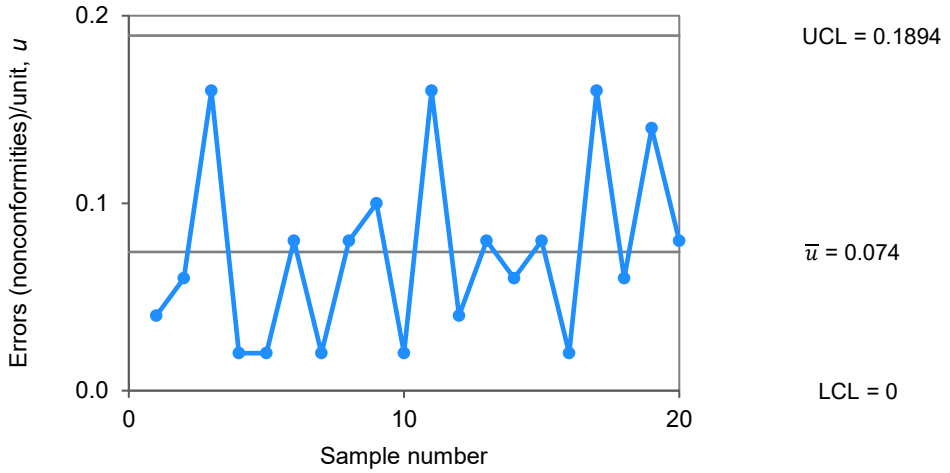


Figure 6-16 The control chart for nonconformities per unit from Minitab for Example 6-4.

The preliminary data do not exhibit lack of statistical control; therefore, the trial control limits given here would be adopted for phase II monitoring of future operations. Once again, note that, although the process is in control, the average number of errors per shipment is high. Action should be taken to improve the supply chain system.

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

6-3.2 Procedures with Variable Sample Size

EXAMPLE 6-5 In a textile finishing plant, dyed cloth is inspected for the occurrence of defects per 50 square meters. The data on ten rolls of cloth are shown in Table 6-11. We will use these data to set up a control chart for nonconformities per unit.

Table 6-11 Occurrence of Nonconformities in Dyed Cloth

Roll Number	Number of Square Meters	Total Number of Nonconformities	Number of Inspection Units in Roll, n	Number of Nonconformities per Inspection Unit
1	500	14	10.0	1.40
2	400	12	8.0	1.50
3	650	20	13.0	1.54
4	500	11	10.0	1.10
5	475	7	9.5	0.74
6	500	10	10.0	1.00
7	600	21	12.0	1.75
8	525	16	10.5	1.52
9	600	19	12.0	1.58
10	625	23	12.5	1.84
		153	107.50	

The center line of the chart should be the average number of nonconformities per inspection unit—that is, the average number of nonconformities per 50 square meters, computed as

$$\overline{u} = \frac{153}{107.5} = 1.42$$

Note that $\overline{u}$ is the ratio of the total number of observed nonconformities to the total number of inspection units.

The control limits on this chart are computed from equation 6-19 with n replaced by n_i. The width of the control limits will vary inversely with n_i, the number of inspection units in the roll. The calculations for the control limits are displayed in Table 6-12. Figure 6-17 plots the control chart constructed by Minitab.

Table 6-12 Calculation of Control Limits, Example 6-5

Roll Number, i	n_i	$UCL = \bar{u} + 3\sqrt{(\bar{u}/n_i)}$	$LCL = \bar{u} - 3\sqrt{(\bar{u}/n_i)}$
1	10.0	2.55	0.29
2	8.0	2.68	0.16
3	13.0	2.41	0.43
4	10.0	2.55	0.29
5	9.5	2.58	0.26
6	10.0	2.55	0.29
7	12.0	2.45	0.39
8	10.5	2.52	0.32
9	12.0	2.45	0.39
10	12.5	2.43	0.41

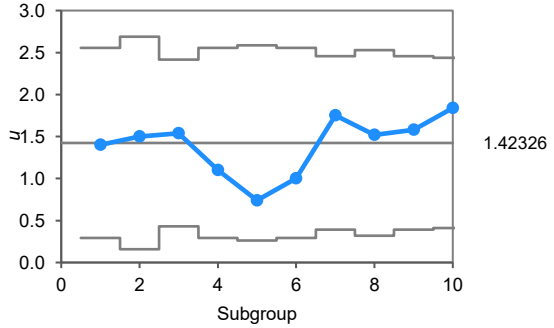


Figure 6-17 Computer-generated (Minitab) control chart for Example 6-5.

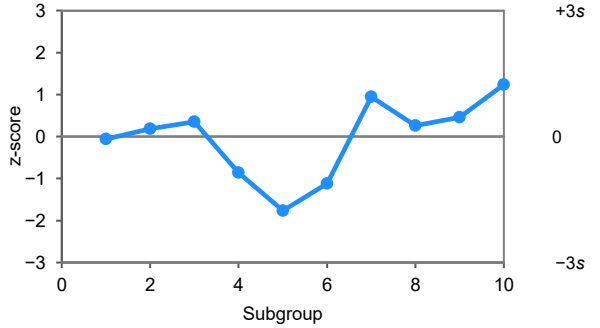


Figure 6-18 Standardized control chart for nonconformities per unit, Example 6-5.

As noted previously, the u chart should always be used when the sample size is variable. The most common implementation involves variable control limits, as illustrated in Example 6-5. There are, however, two other possible approaches:

1. Use control limits based on an average sample size

$$\overline{n} = \sum_{i=1}^{m} n_i / m$$

2. Use a standardized control chart (this is the preferred option).

This second alternative would involve plotting a standardized statistic

$$Z_i = \frac{u_i - \overline{u}}{\sqrt{\frac{\overline{u}}{n_i}}} \tag{6-20}$$

on a control chart with LCL = −3 and UCL = +3 and the center line at zero. This chart is appropriate if tests for runs and other pattern-recognition methods are to be used in conjunction with the chart. Figure 6-18 shows the standardized version of the control chart in Example 6-5. This standardized control chart could also be useful in the short production run situation (see Chapter 9, Section 9-1).

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

6-3.3 Demerit Systems

One possible demerit scheme is defined as follows.

Class A Defects—Very Serious. The unit is either completely unfit for service, or will fail in service in such a manner that cannot be easily corrected in the field, or will cause personal injury or property damage.

Class B Defects—Serious. The unit will possibly suffer a Class A operating failure, or will certainly cause somewhat less serious operating problems, or will certainly have reduced life or increased maintenance cost.

Class C Defects—Moderately Serious. The unit will possibly fail in service, or cause trouble that is less serious than operating failure, or possibly have reduced life or increased maintenance costs, or have a major defect in finish, appearance, or quality of work.

Class D Defects—Minor. The unit will not fail in service but has minor defects in finish, appearance, or quality of work.

Let c_{iA}, c_{iB}, c_{iC}, and c_{iD} represent the number of Class A, Class B, Class C, and Class D defects, respectively, in the ith inspection unit. We assume that each class of defect is independent, and the occurrence of defects in each class is well modeled by a Poisson distribution. Then we define the number of **demerits** in the inspection unit as

$$d_i = 100c_{iA} + 50c_{iB} + 10c_{iC} + c_{iD} \tag{6-21}$$

The demerit weights of Class A—100, Class B—50, Class C—10, and Class D—1 are used fairly widely in practice. However, any reasonable set of weights appropriate for a specific problem may also be used.

Suppose that a sample of n inspection units is used. Then the number of demerits per unit is

$$u_i = \frac{D}{n} \tag{6-22}$$

where $D = \sum_{i=1}^{n} d_i$ is the total number of demerits in all n inspection units.

Since u_i is a linear combination of independent Poisson random variables, the statistics u_i could be plotted on a control chart with the following parameters:

$$\begin{aligned} \text{UCL} &= \overline{u} + 3\hat{\sigma}_u \\ \text{Center line} &= \overline{u} \\ \text{LCL} &= \overline{u} - 3\hat{\sigma}_u \end{aligned} \tag{6-23}$$

where

$$\overline{u} = 100\overline{u}_A + 50\overline{u}_B + 10\overline{u}_C + \overline{u}_D \tag{6-24}$$

and

$$\hat{\sigma}_u = \left[\frac{(100)^2\overline{u}_A + (50)^2\overline{u}_B + (10)^2\overline{u}_C + \overline{u}_D}{n}\right]^{1/2} \tag{6-25}$$

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

6-3.4 The Operating-Characteristic Function

The operating-characteristic (OC) curves for both the c chart and the u chart can be obtained from the Poisson distribution. For the c chart, the OC curve plots the probability of type II error β against the true mean number of defects c. The expression for β is

$$\beta = P\{x < \text{UCL} \mid c\} - P\{x \leq \text{LCL} \mid c\} \tag{6-26}$$

where x is a Poisson random variable with parameter c. Note that if the LCL < 0 the second term on the right-hand side of equation 6-26 should be dropped.

We will generate the OC curve for the c chart in Example 6-3. For this example, since the LCL = 6.48 and the UCL = 33.22, equation 6-26 becomes

$$\beta = P\{x < 33.22 \mid c\} - P\{x \leq 6.48 \mid c\}$$

Since the number of nonconformities must be integer, this is equivalent to

$$\beta = P\{x \leq 33 \mid c\} - P\{x \leq 6 \mid c\}$$

These probabilities are evaluated in Table 6-13. The OC curve is shown in Fig. 6-19.

Table 6-13 Calculation of the OC Curve for a c Chart with UCL = 33.22 and LCL = 6.48

c	$P\{x \leq 33 \mid c\}$	$P\{x \leq 6 \mid c\}$	$\beta = P\{x \leq 33 \mid c\} - P\{x \leq 6 \mid c\}$
1	1.000	0.999	0.001
3	1.000	0.966	0.034
5	1.000	0.762	0.238
7	1.000	0.450	0.550
10	1.000	0.130	0.870
15	0.999	0.008	0.991
20	0.997	0.000	0.997
25	0.950	0.000	0.950
30	0.744	0.000	0.744
33	0.546	0.000	0.546
35	0.410	0.000	0.410
40	0.151	0.000	0.151
45	0.038	0.000	0.038

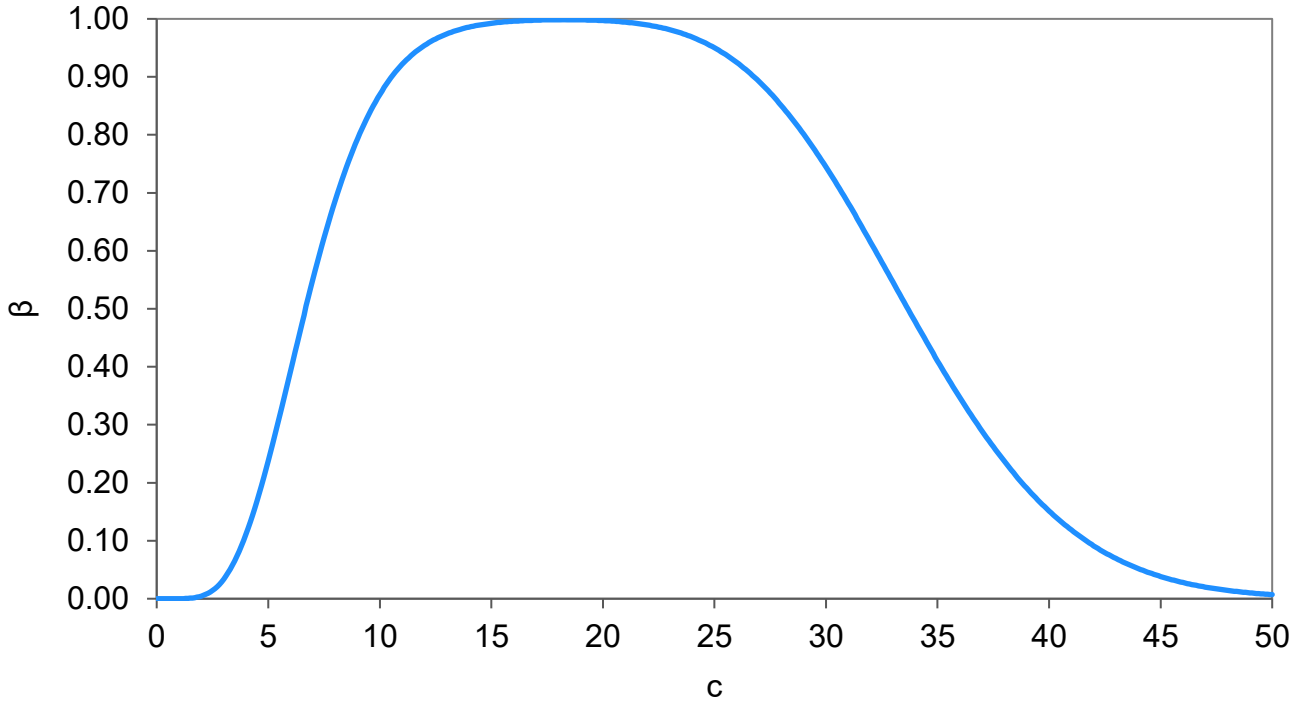


Figure 6-19 OC curve of a *c* chart with LCL = 6.48 and UCL = 33.22.

6-3 CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS)

6-3.5 Dealing with Low Defect Levels

- When defect levels or in general, count rates, in a process become very low—say, under 1000 occurrences per million—there will be very long periods of time between the occurrence of a nonconforming unit. Conventional *c* and *u* charts become ineffective as count rates are driven into the low parts per million (ppm) range.
- The time-between-events control chart has been very effective as a process-control procedure for processes with low defect levels. If the "events" of interest occur according to a Poisson distribution, the probability distribution of the time between events is the exponential distribution.
- Constructing a time-between-events control chart is essentially equivalent to control charting an exponentially distributed variable.

Nelson (1994) has suggested solving this problem by transforming the exponential random variable to a Weibull random variable such that the resulting Weibull distribution is well approximated by the normal distribution. If y represents the original exponential random variable, the appropriate transformation is

$$x = y^{1/3.6} = y^{0.2777} \tag{6-28}$$

One would now construct a control chart on x, assuming that x follows a normal distribution.

EXAMPLE 6-6

A chemical engineer wants to set up a control chart for monitoring the occurrence of failures of an important valve. She has decided to use the number of hours between failures as the variable to monitor. Table 6-14 shows the number of hours between failures for the last 20 failures of this valve. Figure 6-20 is a normal probability plot of the time between failures. Clearly, time between failures is not normally distributed.

Table 6-14 also shows the values of the transformed time between events, computed from equation 6-27. Figure 6-21 is a normal probability plot of the transformed time between failures. Note that the plot indicates that the distribution of this transformed variable is well approximated by the normal.

Figure 6-22 is a control chart for individuals and a moving range control chart for the transformed time between failures. Note that the control charts indicate a state of control, implying that the failure mechanism for this valve is constant. If a process change is made that improves the failure rate (such as a different type of maintenance action), then we would expect to see the mean time between failures get longer. This would result in points plotting above the upper control limit on the individuals control chart in Fig. 6-22.

Table 6-14 Time between Failure Data, Example 6-6

Failure	Time between Failures, y (hr)	Transformed Value of Time between Failures, $x = y^{0.2777}$
1	286	4.80986
2	948	6.70903
3	536	5.72650
4	124	3.81367
5	816	6.43541
6	729	6.23705
7	4	1.46958
8	143	3.96768
9	431	5.39007
10	8	1.78151
11	2837	9.09619
12	596	5.89774
13	81	3.38833
14	227	4.51095
15	603	5.91690
16	492	5.59189
17	1199	7.16124
18	1214	7.18601
19	2831	9.09083
20	96	3.55203

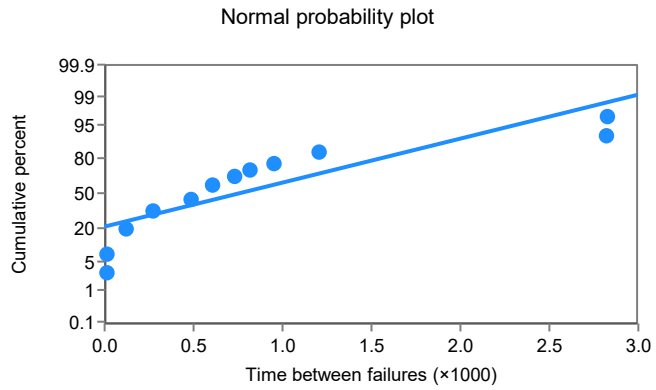


Figure 6-20 Normal probability plot of time between failures, Example 6-6.

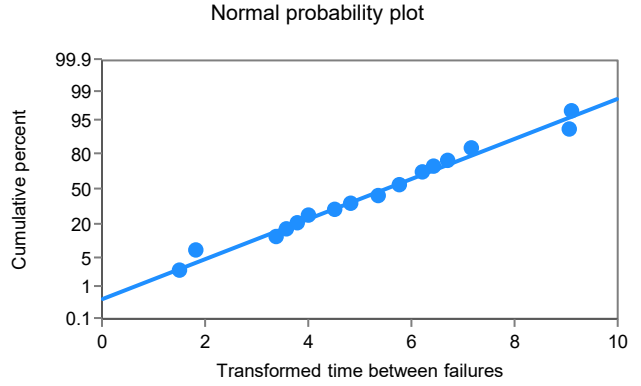


Figure 6-21 Normal probability plot for the transformed failure data.

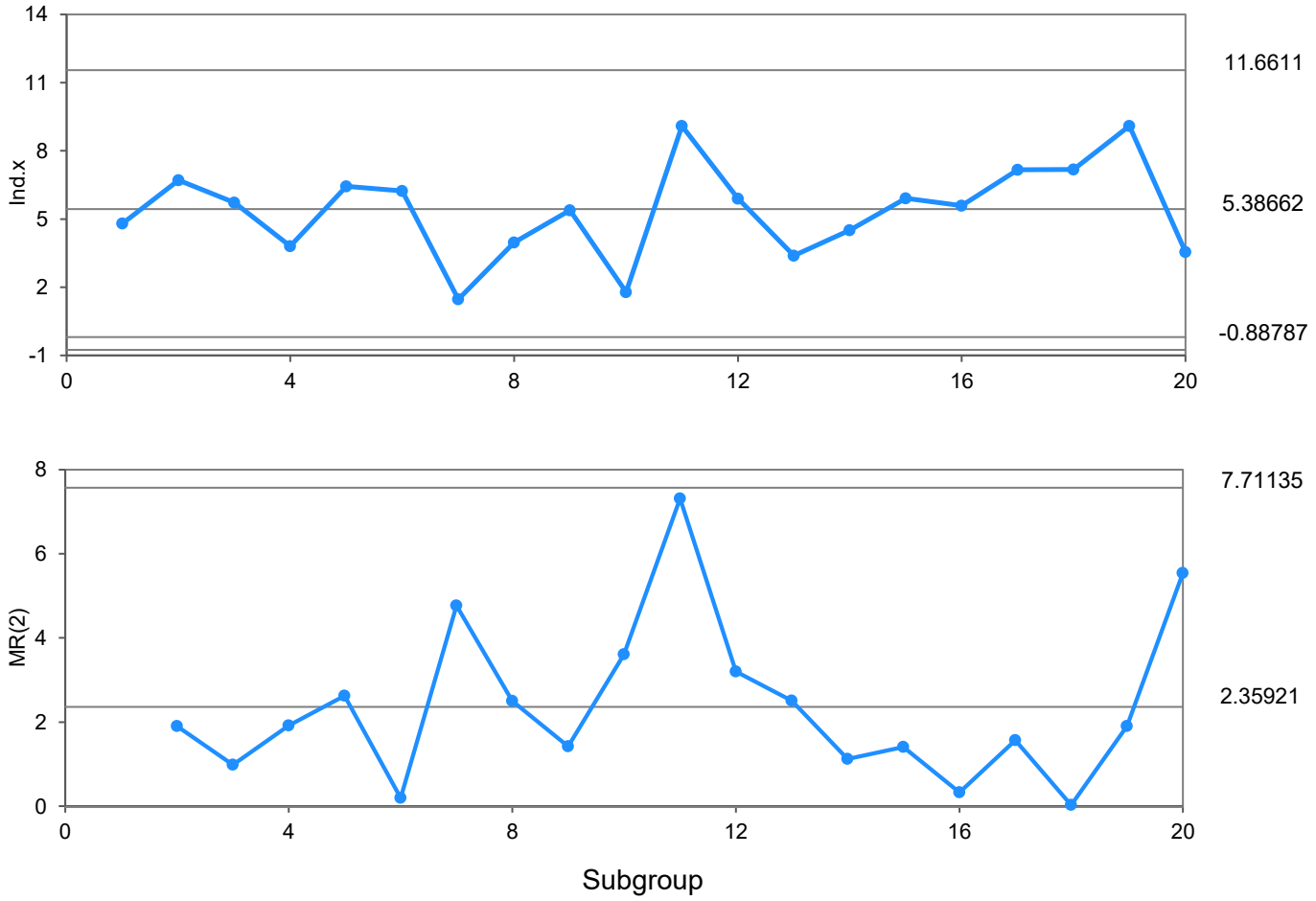


Figure 6-22 Control charts for individuals and moving-range control chart for the transformed time between failures, Example 6-6.

6-5 GUIDELINES FOR IMPLEMENTING CONTROL CHARTS

Almost any process will benefit from SPC, including the use of control charts. In this section, we present some general guidelines helpful in implementing control charts. Specifically, we deal with the following:

1. Determining **which** process characteristics to control
2. Determining **where** the charts should be implemented in the process
3. Choosing the proper **type** of control charts
4. Taking **actions** to improve processes as the result of SPC/control chart analysis
5. Selecting data-collection systems and computer software

The guidelines are applicable to both variables and attributes control charts. Remember, control charts are not only for process surveillance; they should be used as an active, on-line method for reduction of process variability.

Determining Which Characteristics to Control and Where to Put the Control Charts

At the start of a control chart program, it is usually difficult to determine which product or process characteristics should be controlled and at which points in the process to apply control charts. Some useful guidelines follow.

1. At the beginning of a control chart program, control charts should be applied to any product characteristics or manufacturing operations believed to be important. The charts will provide immediate feedback as to whether they are actually needed.
2. The control charts found to be unnecessary should be removed, and others that engineering and operator judgment indicates may be required should be added. More control charts will usually be employed at the beginning than after the process has stabilized.
3. Information on the number and types of control charts on the process should be kept current. It is best to keep separate records on the variables and attributes charts. In general, after the control charts are first installed, we often find that the number of control charts tends to increase rather steadily. After that, it will usually decrease. When the process stabilizes, we typically find that it has the same number of charts from one year to the next. However, they are not necessarily the same charts.

4. If control charts are being used effectively and if new knowledge is being gained about the key process variables, we should find that the number of $\overline{x}$ and R charts increases and the number of attributes control charts decreases.
5. At the beginning of a control chart program there will usually be more attributes control charts, applied to semifinished or finished units near the end of the manufacturing process. As we learn more about the process, these charts will be replaced with $\overline{x}$ and R charts applied *earlier* in the process to the critical parameters and operations that result in nonconformities in the finished product. Generally, **the earlier that process control can be established, the better.** In a complex assembly process, this may imply that process controls need to be implemented at the vendor or supplier level.
6. Control charts are an on-line, process-monitoring procedure. They should be implemented and maintained as close to the work center as possible, so that feedback will be rapid. Furthermore, the process operators and process engineering should have direct responsibility for collecting the process data, maintaining the charts, and interpreting the results. The operators and engineers have the detailed knowledge of the process required to correct process upsets and use the control chart to improve process performance. Microcomputers can speed up the feedback and should be an integral part of any modern, on-line, process-control procedure.
7. The out-of-control-action plan (OCAP) is a vital part of the control chart. Operating and engineering personnel should strive to keep OCAPs up-to-date and valid.

Choosing the Proper Type of Control Chart

A. $\bar{x}$ and R (or $\bar{x}$ and s) charts. Consider using variables control charts in these situations:

1. A new process is coming on stream, or a new product is being manufactured by an existing process.
2. The process has been in operation for some time, but it is chronically in trouble or unable to hold the specified tolerances.
3. The process is in trouble, and the control chart can be useful for diagnostic purposes (troubleshooting).
4. Destructive testing (or other expensive testing procedures) is required.
5. It is desirable to reduce acceptance-sampling or other downstream testing to a minimum when the process can be operated in control.
6. Attributes control charts have been used, but the process is either out of control or in control but the yield is unacceptable.
7. There are very tight specifications, overlapping assembly tolerances, or other difficult manufacturing problems.
8. The operator must decide whether or not to adjust the process, or when a setup must be evaluated.
9. A change in product specifications is desired.
10. Process stability and capability must be continually demonstrated, such as in regulated industries.

B. Attributes Charts (p charts, c charts, and u charts). Consider using attributes control charts in these situations:

1. Operators control the assignable causes, and it is necessary to reduce process fallout.
2. The process is a complex assembly operation and product quality is measured in terms of the occurrence of nonconformities, successful or unsuccessful product function, and so forth. (Examples include computers, office automation equipment, automobiles, and the major subsystems of these products.)
3. Process control is necessary, but measurement data cannot be obtained.
4. A historical summary of process performance is necessary. Attributes control charts, such as p charts, c charts, and u charts, are very effective for summarizing information about the process for management review.
5. Remember that attributes charts are generally inferior to charts for variables. Always use $\overline{x}$ and R or $\overline{x}$ and s charts whenever possible.

C. Control Charts for Individuals. Consider using the control chart for individuals in conjunction with a moving-range chart in these situations:

1. It is inconvenient or impossible to obtain more than one measurement per sample, or repeat measurements will only differ by laboratory or analysis error. Examples often occur in chemical processes.
2. Automated testing and inspection technology allow measurement of every unit produced. In these cases, also consider the cumulative sum control chart and the exponentially weighted moving average control chart discussed in Chapter 7.
3. The data become available very slowly, and waiting for a larger sample will be impractical or make the control procedure too slow to react to problems. This often happens in nonproduct situations; for example, accounting data may become available only monthly.
4. Generally, once we are in phase II, individuals charts have poor performance in shift detection and can be very sensitive to departures from normality. Always use the EWMA and cusum charts of Chapter 8 in phase II instead of individuals charts whenever possible.

Actions Taken to Improve the Process

Figure 6-27 gives the answers to two questions: "Is the process in control?" and "Is the process capable (in the sense of the previous paragraph)?" Each of the four cells in the figure contains some recommended courses of action that depend on the answers to these two questions.

IS THE PROCESS IN CONTROL?	IS THE PROCESS CAPABLE? Yes	IS THE PROCESS CAPABLE? No
Yes	SPC	SPC Experimental design Investigate specifications Change process
No	SPC	SPC Experimental design Investigate specifications Change process

Figure 6-27 Actions taken to improve a process.

The lower two boxes in Fig. 6-27 deal with the case of an out-of-control process. The southeast corner presents the case of a process that is out of control and not capable. (Remember our nontechnical use of the term *capability*.) The actions recommended here are identical to those for the box in the northeast corner, except that SPC would be expected to yield fairly rapid results now, because the control charts should be identifying the presence of assignable causes. The other methods of attack will warrant consideration and use in many cases, however. Finally, the southwest corner treats the case of a process that exhibits lack of statistical control but does not produce a meaningful number of defectives because the specifications are very wide. SPC methods should still be used to establish control and reduce variability in this case, for the following reasons:

1. Specifications can change without notice.
2. The customer may require both **control** and **capability.**
3. The fact that the process experiences assignable causes implies that unknown forces are at work; these unknown forces could result in poor capability in the near future.

IMPORTANT TERMS AND CONCEPTS

Attribute data

Average run length for attribute control charts

Cause-and-effect diagram

Choice between attributes and variables data

Control chart for defects or nonconformities per unit or *u* chart

Control chart for fraction nonconforming or *p* chart

Control chart for nonconformities or *c* chart

Control chart for number nonconforming or *np* chart

Defect

Defective

Demerit systems for attribute data

Design of attributes control charts

Fraction defective

Fraction nonconforming

Nonconformities

Operating characteristic curve for the *c* and *u* charts

Operating characteristic curve for the *p* chart

Pareto chart

Standardized control charts

Time between occurrence control charts

Variable sample size for attributes control chart

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LEARNING OBJECTIVES

1. Understand the statistical basis of attributes control charts
2. Know how to design attributes control charts
3. Know how to set up and use the p chart for fraction nonconforming
4. Know how to set up and use the np control chart for the number of nonconforming items
5. Know how to set up and use the c control chart for defects
6. Know how to set up and use the u control chart for defects per unit
7. Use attributes control charts with variable sample size
8. Understand the advantages and disadvantages of attributes versus variables control charts
9. Understand the rational subgroup concept for attributes control charts
10. Determine the average run length for attributes control charts

Sample size: When p is very small, we should choose n large enough so that the probability of finding at least one defective item in the sample is high enough; otherwise it would cause the unreasonable situation that the constructed C.C. rejects as soon as the sample contains one defective item.

e.g. If $p = 0.01$, $n = 8$, then

$$\text{UCL} = p + 3\sqrt{\frac{p(1-p)}{n}} = 0.1155$$

then if $\hat{p} = 1/8 = 0.125 > \text{UCL}$,

→ a single defective item indicates out of control !!

Suggestion I: If D = the number of defective items, then we can require

$$P(D \geq 1) = 1 - P(D = 0) = 1 - \binom{n}{0} p^0 (1-p)^n = 1 - (1-p)^n \geq r$$

e.g. We may take $r = 0.95$; then when $p = 0.01$,

$$1 - (1 - 0.01)^n \geq 0.95 \Rightarrow n \geq \frac{\ln 0.05}{\ln 0.99} \simeq 298$$

Alternatively, the Poisson approximation to the Binomial can be used, looking up the Poisson table.

(But in this case it must be required that $\lambda = np \geq 3.00$,

so when $p = 0.01$, $n \geq 300$)

Suggestion II: Duncan (1974)

n must be large enough so that, when the process shifts by a certain specified amount, the probability that we can detect this shift is about 50%.

Duncan (1974): The sample size must be chosen large enough so that the control chart has about a 50% probability of detecting a change in the nonconforming rate. Suppose the in-control nonconforming rate is p_1, and we want the control chart to detect the change with probability 50% when the nonconforming rate increases to p_2. Then the normal distribution can be used to approximate the binomial distribution, and n is chosen so that p_2 falls exactly on the UCL (according to the normal distribution, when the mean after the shift falls on the control limit, the probability of its being detected is 50%). Therefore, if $\delta = p_2 - p_1$, n must satisfy

$$\delta = \kappa\sqrt{\frac{p(1-p)}{n}}$$

$$\text{Therefore } n = \left(\frac{\kappa}{\delta}\right)^2 p(1-p)$$

That is, choose n so that its upper control limit equals the nonconforming rate when out of control.

Continuing the example above,

$$p = 0.01,\ \delta = 0.05 - 0.01 = 0.04,$$

take $k = 3$, then $n = \left(\frac{3}{0.04}\right)^2 (0.01)(0.99) = 56$

If *p* when in control is very small, *n* can also be chosen so that the lower limit of the C.C. is greater than 0; this helps to detect some unusually low nonconforming rates (this phenomenon is usually caused by inspection error, "e.g. an inexperienced inspector", rather than representing an improvement in quality). That is,

$$\text{LCL} = p - k\sqrt{\frac{p(1-p)}{n}} > 0 \;\Rightarrow\; n > \left(\frac{1-p}{p}\right)k^2$$

e.g. $p = 0.05,\ k = 3,\ n > 171$

Process and Measurement System Capability Analysis

Chapter 7. Process and Measurement System Capability Analysis

Process Capability

1. The natural or inherent variability in a critical-to-quality characteristic at a specified time; that is, "instantaneous" variability

2. The variability in a critical-to-quality characteristic over time

Natural tolerance limits are defined as follows:

$$\text{UNTL} = \mu + 3\sigma$$
$$\text{LNTL} = \mu - 3\sigma$$

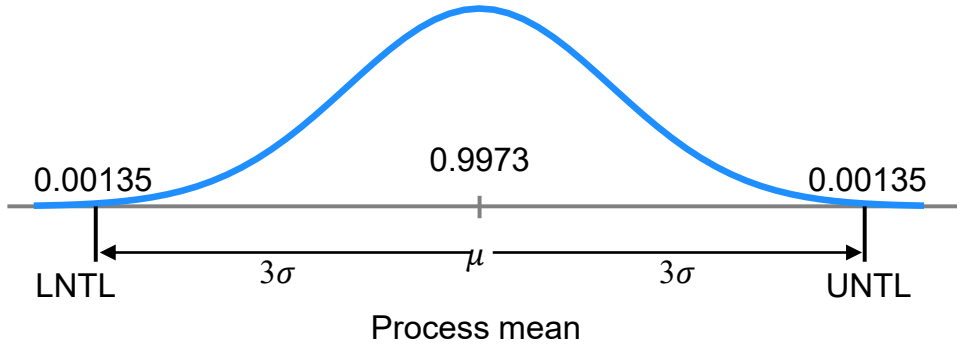


Figure 7-1 Upper and lower natural tolerance limits in the normal distribution.

Uses of process capability

1. Predicting how well the process will hold the tolerances
2. Assisting product developers/designers in selecting or modifying a process
3. Assisting in establishing an interval between sampling for process monitoring
4. Specifying performance requirements for new equipment
5. Selecting between competing suppliers and other aspects of supply chain management
6. Planning the sequence of production processes when there is an interactive effect of processes on tolerances
7. Reducing the variability in a manufacturing process

Reasons for Poor Process Capability



Figure 7-3 Some reasons for poor process capability. (*a*) Poor process centering. (*b*) Excess process variability.

Process may have good potential capability

Data collection steps:

1. Choose the machine or machines to be used. If the results based on one (or a few) machines are to be extended to a larger population of machines, the machine selected should be representative of those in the population. Furthermore, if the machine has multiple workstations or heads, it may be important to collect the data so that head-to-head variability can be isolated. This may imply that designed experiments should be used.

2. Select the process operating conditions. Carefully define conditions, such as cutting speeds, feed rates, and temperatures, for future reference. It may be important to study the effects of varying these factors on process capability.

3. Select a representative operator. In some studies, it may be important to estimate *operator* variability. In these cases, the operators should be selected at random from the population of operators.

4. Carefully monitor the data-collection process, and record the time order in which each unit is produced.

Example 7-1, Glass Container Data

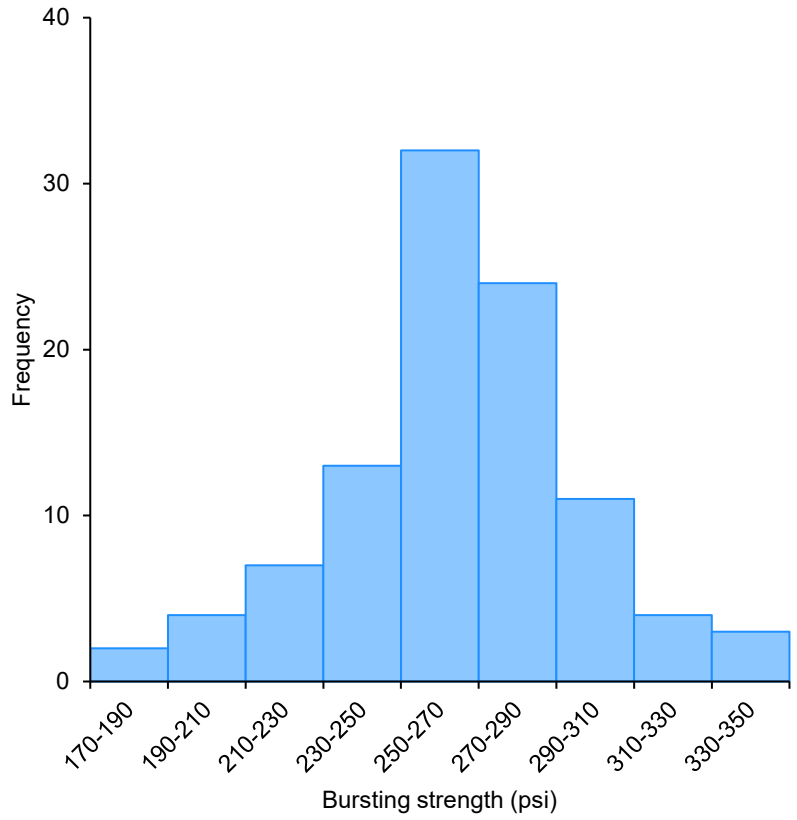


Figure 7-2 Histogram for the bursting-strength data.

Table 7-1 Bursting Strengths for 100 Glass Containers

265	197	346	280	265	200	221	265	261	278
205	286	317	242	254	235	176	262	248	250
263	274	242	260	281	246	248	271	260	265
307	243	258	321	294	328	263	245	274	270
220	231	276	228	223	296	231	301	337	298
268	267	300	250	260	276	334	280	250	257
260	281	208	299	308	264	280	274	278	210
234	265	187	258	235	269	265	253	254	280
299	214	264	267	283	235	272	287	274	269
215	318	271	293	277	290	283	258	275	251

EXAMPLE 7-1

To illustrate the use of a histogram to estimate process capability, consider Fig. 7-2, which presents a histogram of the bursting strength of 100 glass containers. The data are shown in Table 7-1. Analysis of the 100 observations gives

$$\overline{x} = 264.06 \qquad s = 32.02$$

Consequently, the process capability would be estimated as

$$\overline{x} \pm 3s$$

or

$$264.06 \pm 3(32.02) \simeq 264 \pm 96 \text{ psi}$$

Furthermore, the shape of the histogram implies that the distribution of bursting strength is approximately normal. Thus, we can estimate that approximately 99.73% of the bottles manufactured by this process will burst between 168 and 360 psi. Note that we can estimate process capability *independent of the specifications on bursting strength.*

Probability Plotting

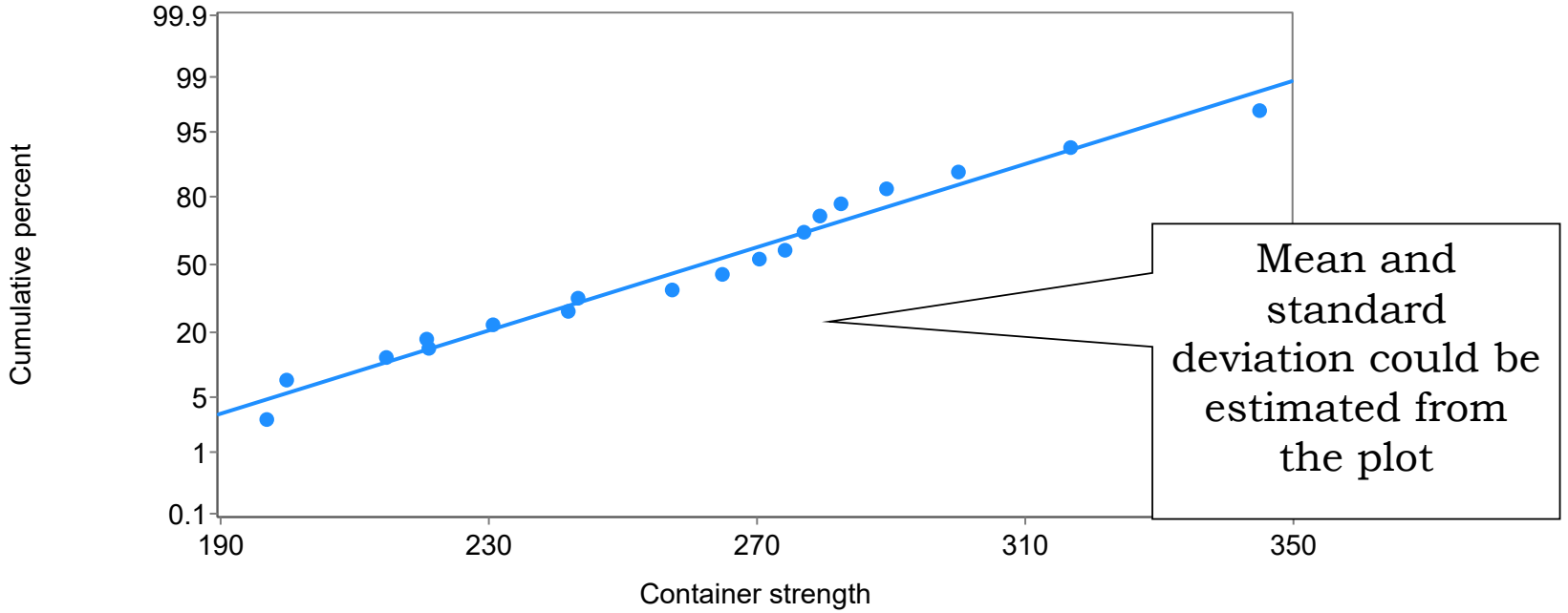


Figure 7-4 Normal probability plot of the container-strength data.

- The distribution may not be normal; other types of probability plots can be useful in determining the appropriate distribution.

Process Capability Ratios

7-3 PROCESS CAPABILITY RATIOS

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \qquad (7\text{-}4)$$

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} \qquad (7\text{-}5)$$

Example 5-1: USL = 1.00 microns, LSL = 2.00 microns

$$\sigma = \frac{\overline{R}}{d_2} = 0.1398$$

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} = \frac{2.00 - 1.00}{6(0.1398)} = 1.192$$

Practical interpretation
PCR = proportion of tolerance interval used by process

$$P = \left(\frac{1}{C_p}\right)100 \qquad (7\text{-}6)$$

For the hard bake process:

$$P = \left(\frac{1}{1.192}\right)100 = 83.89$$

One-Sided PCR

$$C_{pu} = \frac{\text{USL} - \mu}{3\sigma} \quad \text{(upper specification only)} \quad (7-7)$$

$$C_{pl} = \frac{\mu - \text{LSL}}{3\sigma} \quad \text{(lower specification only)} \quad (7-8)$$

Table 7-2 Values of the Process Capability Ratio (C_p) and Associated Process Fallout for a Normally Distributed Process (in Defective ppm) That Is in Statistical Control

PCR	Process Fallout (in defective ppm)	
	One-Sided Specifications	Two-Sided Specifications
0.25	226,628	453,255
0.50	66,807	133,614
0.60	35,931	71,861
0.70	17,865	35,729
0.80	8,198	16,395
0.90	3,467	6,934
1.00	1,350	2,700
1.10	484	967
1.20	159	318
1.30	48	96
1.40	14	27
1.50	4	7
1.60	1	2
1.70	0.17	0.34
1.80	0.03	0.06
2.00	0.0009	0.0018

Table 7-3 Recommended Minimum Values of the Process Capability Ratio

	Two-Sided Specifications	One-Sided Specifications
Existing processes	1.33	1.25
New processes	1.50	1.45
Safety, strength, or critical parameter, existing process	1.50	1.45
Safety, strength, or critical parameter, new process	1.67	1.60

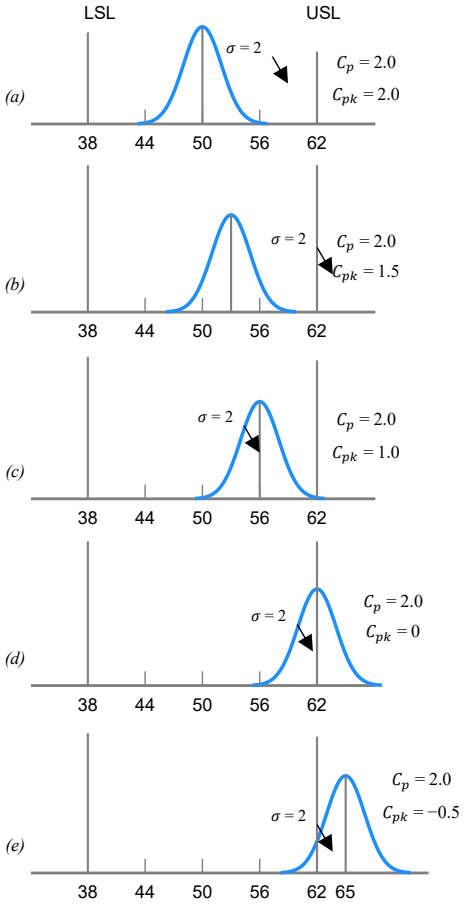


Figure 7-8 Relationship of C_p and C_{pk}.

- C_p does not take process centering into account
- It is a measure of *potential* capability, not *actual* capability

Measure of Actual Capability

$$C_{pk} = \min\left(C_{pu}, C_{pl}\right) \tag{7-9}$$

Note that C_{pk} is simply the one-sided PCR for the specification limit nearest to the

C_{pk}: one-sided PCR for specification limit nearest to process average

$$C_{pk} = \min\left(C_{pu}, C_{pl}\right)$$

$$= \min\left(C_{pu} = \frac{\text{USL} - \mu}{3\sigma},\ C_{pl} = \frac{\mu - \text{LSL}}{3\sigma}\right)$$

$$= \min\left(C_{pu} = \frac{62 - 53}{3(2)} = 1.5,\ C_{pl} = \frac{53 - 38}{3(2)} = 2.5\right)$$

$$= 1.5$$

Normality and Process Capability Ratios

- The assumption of normality is critical to the interpretation of these ratios.
- For non-normal data, options are:
  1. Transform non-normal data to normal.
  2. Extend the usual definitions of PCRs to handle non-normal data.
  3. Modify the definitions of PCRs for general families of distributions.

- Confidence intervals are an important way to express the information in a PCR

$$\hat{C}_p\sqrt{\frac{\chi^2_{1-\alpha/2,n-1}}{n-1}} \le C_p \le \hat{C}_p\sqrt{\frac{\chi^2_{\alpha/2,n-1}}{n-1}} \qquad (7\text{-}20)$$

EXAMPLE 7-4

Suppose that a stable process has upper and lower specifications at USL = 62 and LSL = 38. A sample of size n = 20 from this process reveals that the process mean is centered approximately at the midpoint of the specification interval and that the sample standard deviation s = 1.75. Therefore, a point estimate of C_p is

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6s} = \frac{62 - 38}{6(1.75)} = 2.29$$

The 95% confidence interval on C_p is found from equation 7–20 as follows:

$$\hat{C}_p \sqrt{\frac{\chi^2_{1-0.025,n-1}}{n-1}} \le C_p \le \hat{C}_p \sqrt{\frac{\chi^2_{0.025,n-1}}{n-1}}$$

$$2.29\sqrt{\frac{8.91}{19}} \le C_p \le 2.29\sqrt{\frac{32.85}{19}}$$

$$1.57 \le C_p \le 3.01$$

Confidence interval is wide because the sample size is small

where $\chi^2_{0.975,19}$ = 8.91 and $\chi^2_{0.025,19}$ = 32.85 were taken from Appendix Table III.

EXAMPLE 7-5

A sample of size n = 20 from a stable process is used to estimate C_{pk}, with the result that $\hat{C}_{pk}$ = 1.33. Using equation 7–21, an approximate 95% confidence interval on C_{pk} is

$$\hat{C}_{pk}\left[1 - Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}}\right] \le C_{pk} \le \hat{C}_{pk}\left[1 + Z_{\alpha/2}\sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}}\right]$$

$$1.33\left[1 - 1.96\sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}}\right] \le C_{pk} \le 1.33\left[1 + 1.96\sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}}\right]$$

or

$$0.88 \le C_{pk} \le 1.78$$

Confidence interval is wide because the sample size is small

Process Performance Index: $\hat{P}_p = \dfrac{\text{USL} - \text{LSL}}{6s}$

Use only when the process is not in control.

If the process is normally distributed and in control:

$$\hat{P}_p = \hat{C}_p \quad \text{and} \quad \hat{P}_{pk} = \hat{C}_{pk} \quad \text{because} \quad \hat{\sigma} \approx \frac{\overline{R}}{d_2}$$

Process Capability Analysis Using Control Chart

7-4 PROCESS CAPABILITY ANALYSIS USING A CONTROL CHART

Table 7-5 Glass Container Strength Data (psi)

Sample	Data					$\bar{x}$	R
1	265	205	263	307	220	252.0	102
2	268	260	234	299	215	255.2	84
3	197	286	274	243	231	246.2	89
4	267	281	265	214	318	269.0	104
5	346	317	242	258	276	287.8	104
6	300	208	187	264	271	246.0	113
7	280	242	260	321	228	266.2	93
8	250	299	258	267	293	273.4	49
9	265	254	281	294	223	263.4	71
10	260	308	235	283	277	272.6	73
11	200	235	246	328	296	261.0	128
12	276	264	269	235	290	266.8	55
13	221	176	248	263	231	227.8	87
14	334	280	265	272	283	286.8	69
15	265	262	271	245	301	268.8	56
16	280	274	253	287	258	270.4	34
17	261	248	260	274	337	276.0	89
18	250	278	254	274	275	266.2	28
19	278	250	265	270	298	272.2	48
20	257	210	280	269	251	253.4	70
						$\bar{\bar{x}} = 264.06$	$\bar{R} = 77.3$

R Chart

$$\text{Center line} = \bar{R} = 77.3$$
$$\text{UCL} = D_4\bar{R} = (2.115)(77.3) = 163.49$$
$$\text{LCL} = D_3\bar{R} = (0)(77.3) = 0$$

$\bar{x}$ Chart

$$\text{Center line} = \bar{\bar{x}} = 264.06$$
$$\text{UCL} = \bar{\bar{x}} + A_2\bar{R} = 264.06 + (0.577)(77.3) = 308.66$$
$$\text{LCL} = \bar{\bar{x}} - A_2\bar{R} = 264.06 - (0.577)(77.3) = 219.46$$

- Specifications are not needed to estimate parameters.

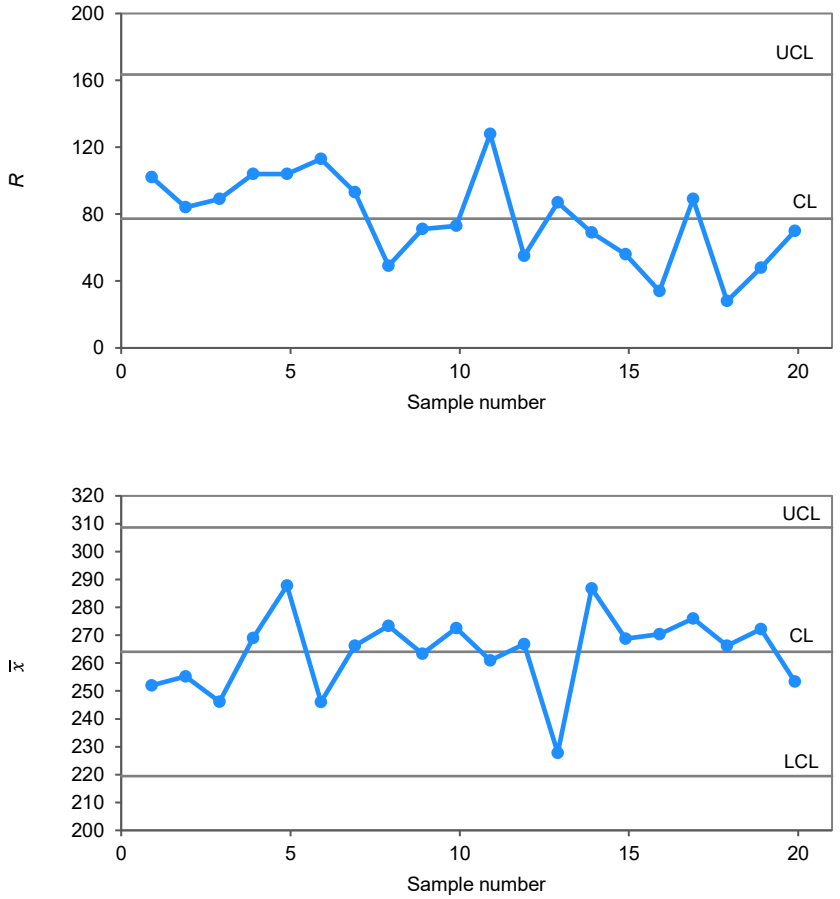


Figure 7-12 $\bar{x}$ and R charts for the bottle-strength data.

$$\hat{\mu} = \bar{\bar{x}} = 264.06$$

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{77.3}{2.326} = 33.23$$

Since LSL = 200

$$\hat{C}_{pl} = \frac{\hat{\mu} - \text{LSL}}{3\hat{\sigma}} = \frac{264.06 - 200}{3(33.23)} = 0.64$$

Process Capability Analysis Using Designed Experiments

7-5 PROCESS CAPABILITY ANALYSIS USING DESIGNED EXPERIMENTS

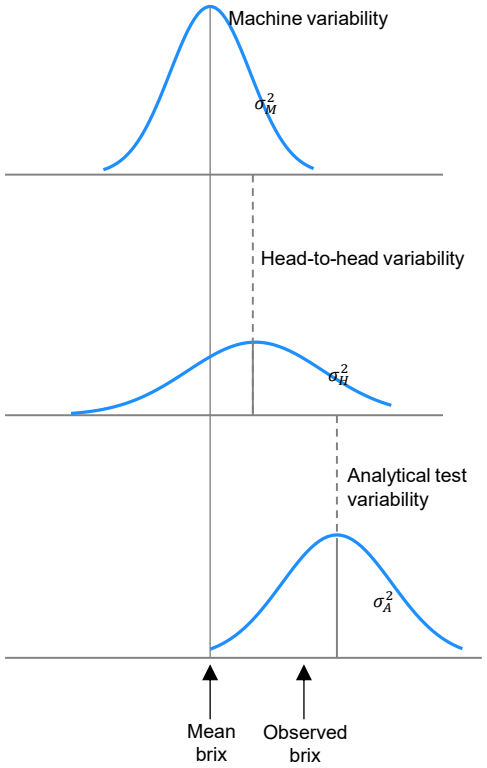


Figure 7-13 Sources of variability in the bottling line example.

Gauge and Measurement System Capability

7-6 GAUGE AND MEASUREMENT SYSTEM CAPABILITY STUDIES

1. Determine how much of the total observed variability is due to the gauge or instrument

2. Isolate the components of variability in the measurement system

3. Assess whether the instrument or gauge is capable (that is, is it suitable for the intended application)

Simple model:

A s

y: total observed measurement -23)

x: true value of measurement

wh it

ε: measurement error

of p

ind ge,

respectively. The variance of the total observed measurement, y, is then

x and ε are independent. $x \sim N(\mu, \sigma_P^2)$ and $\varepsilon \sim N(0, \sigma_{\text{Gauge}}^2)$

$$\sigma_{\text{Total}}^2 = \sigma_P^2 + \sigma_{\text{Gauge}}^2 \quad (7\text{-}24)$$

EXAMPLE 7-7 Measuring Gauge Capability

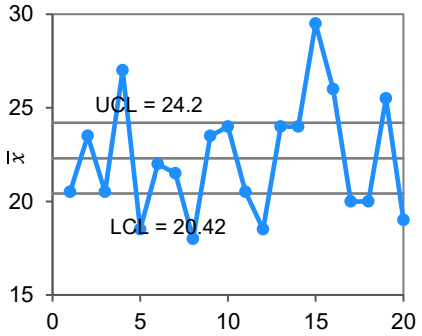
An instrument is to be used as part of a proposed SPC implementation. The quality-improvement team involved in designing the SPC system would like to get an assessment of gauge capability. Twenty units of the product are obtained, and the process operator who will actually take the measurements for the control chart uses the instrument to measure each unit of product twice. The data are shown in Table 7-6.

Figure 7-14 shows the $\overline{x}$ and R charts for these data. Note that the $\overline{x}$ chart exhibits many out-of-control points. This is to be expected, because in this situation the $\overline{x}$ chart has an interpretation that is somewhat different from the usual interpretation. The $\overline{x}$ chart in this example shows the discriminating power of the instrument—literally, the ability of the gauge to distinguish between units of product. The R chart directly shows the magnitude of measurement error, or the gauge capability. The R values represent the difference between measurements made on the same unit using the same instrument. In this example, the R chart is in control. This indicates that the operator is having no difficulty in making consistent measurements. Out-of-control points on the R chart could indicate that the operator is having difficulty using the instrument.

Table 7-6 Parts Measurement Data

Part Number	Measurements		$\bar{x}$	R
	1	2		
1	21	20	20.5	1
2	24	23	23.5	1
3	20	21	20.5	1
4	27	27	27.0	0
5	19	18	18.5	1
6	23	21	22.0	2
7	22	21	21.5	1
8	19	17	18.0	2
9	24	23	23.5	1
10	25	23	24.0	2
11	21	20	20.5	1
12	18	19	18.5	1
13	23	25	24.0	2
14	24	24	24.0	0
15	29	30	29.5	1
16	26	26	26.0	0
17	20	20	20.0	0
18	19	21	20.0	2
19	25	26	25.5	1
20	19	19	19.0	0
			$\bar{\bar{x}} = 22.3$	$\bar{R} = 1.0$

$$\hat{\sigma}_{\text{Gauge}} = \frac{\bar{R}}{d_2} = \frac{1.0}{1.128} = 0.887$$



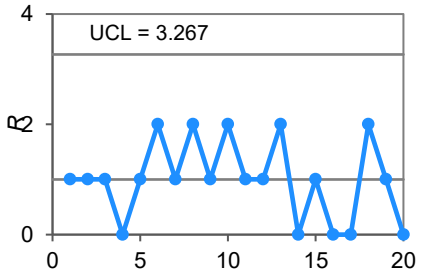


Figure 7-14 Control charts for the gauge capability analysis in Example 7-7.

$$P/T = \frac{k\hat{\sigma}_{\text{Gauge}}}{\text{USL} - \text{LSL}} \qquad (7\text{-}25)$$

$k = 5.15 \rightarrow$ number of standard deviation between 95% tolerance interval Containing 99% of normal population.
$k = 6 \rightarrow$ number of standard deviations between natural tolerance limit of normal population.
For Example 7-7, USL = 60 and LSL = 5. With $k = 6$

$$\widehat{P/T} = \frac{6(0.887)}{60 - 5} = \frac{5.32}{55} = 0.097$$

$P/T \leq 0.1$ is taken as appropriate.

Estimating Variances

$$\hat{\sigma}^2_{\text{Total}} = s^2 = (3.17)^2 = 10.05$$

$$\sigma^2_{\text{Total}} = \sigma^2_P + \sigma^2_{\text{Gauge}}$$

$$\hat{\sigma}^2_P = \hat{\sigma}^2_{\text{Total}} - \hat{\sigma}^2_{\text{Gauge}} = 10.05 - 0.79 = 9.26$$

$\longrightarrow$

$$\hat{\sigma}_P = \sqrt{9.26} = 3.04$$

Signal to Noise Ration (SNR)

Another measure of measurement system adequacy is defined by the AIAG (1995) [note that there is also an updated edition of this manual, AIAG (2002)] as the signal-to-noise ratio (SNR):

$$\text{SNR} = \sqrt{\frac{2\rho_P}{1-\rho_P}} \tag{7-28}$$

SNR: number of distinct levels that can be reliably obtained from measurements. *SNR* should be ≥ 5

$$\widehat{\text{SNR}} = \sqrt{\frac{2\hat{\rho}_P}{1-\hat{\rho}_P}} = \sqrt{\frac{2(0.9214)}{1-0.9214}} = 4.84$$

The gauge (with estimated *SNR* < 5) is not capable according to *SNR* criterion.

Discrimination Ratio

$$\mathrm{DR} = \frac{1 + \rho_P}{1 - \rho_P} \tag{7-29}$$

DR > 4 for capable gauges.
For example, 7-7:

$$\widehat{\mathrm{DR}} = \frac{1 + \hat{\rho}_P}{1 - \hat{\rho}_P} = \frac{1 + 0.9214}{1 - 0.9214} = 24.45$$

The gauge is capable according *to DR* criterion.

Clearly by this measure, the gauge is capable.

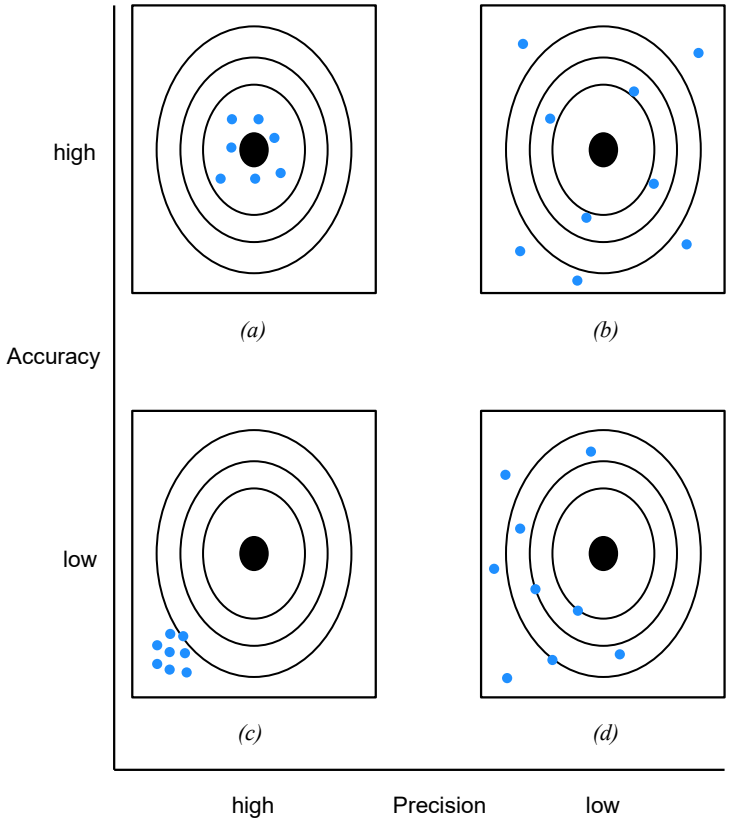


Figure 7-15 The concepts of accuracy and precision. (*a*) The gauge is accurate and precise. (*b*) The gauge is accurate but not precise. (*c*) The gauge is not accurate but it is precise. (*d*) The gauge is neither accurate nor precise.

Gauge R&R Studies

$$\sigma^2_{\text{Measurement Error}} = \sigma^2_{\text{Gauge}} = \sigma^2_{\text{Repeatability}} + \sigma^2_{\text{Reproducibility}} \quad (7\text{-}30)$$

- Usually conducted with a factorial experiment (p: number of randomly selected parts, o: number of randomly selected operators, n: number of times each operator measures each part)

If there are a randomly selected parts and b randomly selected operators, and each operator measures every part n times, then the measurements (i = part, j = operator, k = measurement) could be represented by the model

$$y_{ijk} = \mu + P_i + O_j + (PO)_{ij} + \varepsilon_{ijk} \qquad \begin{cases} i = 1,2,\dots,p \\ j = 1,2,\dots,o \\ k = 1,2,\dots,n \end{cases} \quad (7\text{-}30)$$

$P_i \sim N(0, \sigma_P^2)$: effects of parts
$O_j \sim N(0, \sigma_O^2)$: effects of operators
$(PO)_{ij} \sim N(0, \sigma_{PO}^2)$: joint effects of parts and operators
$\varepsilon_{ijk} \sim N(0, \sigma^2)$: effects of operators

$$V(y_{ijk}) = \sigma_P^2 + \sigma_O^2 + \sigma_{PO}^2 + \sigma^2 \quad (7\text{-}31)$$

Table 7-7 Thermal Impedance Data (°C/W × 100) for the Gauge R & R Experiment

Part Number	Inspector 1			Inspector 2			Inspector 3		
	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3
1	37	38	37	41	41	40	41	42	41
2	42	41	43	42	42	42	43	42	43
3	30	31	31	31	31	31	29	30	28
4	42	43	42	43	43	43	42	42	42
5	28	30	29	29	30	29	31	29	29
6	42	42	43	45	45	45	44	46	45
7	25	26	27	28	28	30	29	27	27
8	40	40	40	43	42	42	43	43	41
9	25	25	25	27	29	28	26	26	26
10	35	34	34	35	35	34	35	34	35

- ANOVA methods are used to conduct the R&R analysis.

$$SS_{\text{Total}} = SS_{\text{Parts}} + SS_{\text{Operators}} + SS_{P\times O} + SS_{\text{Error}} \qquad (7\text{-}32)$$

$$MS_P = \frac{SS_{\text{Parts}}}{p-1}$$

$$MS_O = \frac{SS_{\text{Operators}}}{o-1}$$

$$MS_{PO} = \frac{SS_{P\times O}}{(p-1)(o-1)}$$

$$MS_E = \frac{SS_{\text{Error}}}{po(n-1)}$$

$$E(MS_P) = \sigma^2 + n\sigma_{PO}^2 + bn\sigma_P^2$$

$$E(MS_O) = \sigma^2 + n\sigma_{PO}^2 + an\sigma_O^2$$

$$E(MS_{PO}) = \sigma^2 + n\sigma_{PO}^2$$

$$\hat{\sigma}^2 = MS_E$$

$$\hat{\sigma}_{PO}^2 = \frac{MS_{PO} - MS_E}{n}$$

$$\hat{\sigma}_O^2 = \frac{MS_O - MS_{PO}}{pn}$$

$$\hat{\sigma}_P^2 = \frac{MS_P - MS_{PO}}{on}$$

Table 7-8 ANOVA: Thermal Impedance versus Part Number, Operator

Factor	Type	Levels	Values
Part Num	random	10	1 2 3 4 5 6 7 8 9 10
Operator	random	3	1 2 3

Analysis of Variance for Thermal

Source	DF	SS	MS	F	P
Part Num	9	3935.96	437.33	162.27	0.000
Operator	2	39.27	19.63	7.28	0.005
Part Num*Operator	18	48.51	2.70	5.27	0.000
Error	60	30.67	0.51		
Total	89	4054.40			

Source	Variance component	Error term	Expected Mean Square for Each Term (using unrestricted model)
1 Part Num	48.2926	3	(4) + 3(3) + 9(1)
2 Operator	0.5646	3	(4) + 3(3) + 30(2)
3 Part Num*Operator	0.7280	4	(4) + 3(3)
4 Error	0.5111		(4)

$$\hat{\sigma}_P^2 = \frac{437.33 - 2.70}{(3)(3)} = 48.29$$

$$\hat{\sigma}_O^2 = \frac{19.63 - 2.70}{(10)(3)} = 0.56$$

$$\hat{\sigma}_{PO}^2 = \frac{2.70 - 0.51}{3} = 0.73$$

$$\sigma^2 = 0.51$$

- Negative estimates of a variance component would lead to filling a reduced model, such as, for example:

$$y_{ijk} = \mu + P_i + O_j + \varepsilon_{ijk}$$

Typically we think of σ^2 as the **repeatability** variance component, and the gauge

reσ2: repeatability variance component

Therefore $\longrightarrow$

$\longrightarrow$ $\sigma^2_{Gauge} = \sigma^2_{Reproducibility} + \sigma^2_{Repeatability}$

$\hat{\sigma}^2_{Gauge} = \hat{\sigma}^2 + \hat{\sigma}^2_O + \hat{\sigma}^2_{PO}$

For the example: $= 0.51 + 0.56 + 0.73$

$= 1.80$

cations on this power module are LSL = 18 and USL = 58.
he gauge is estimated as

With LSL = 18 and USL = 58:

This gauge is not capable since $P/T > 0.1$.

Linear Combinations

For $x_1, x_2, \ldots, x_n$, assume $x_i \sim N(\mu_i, \sigma_i^2)$ and independent from each other.

Let $y = a_1x_1 + a_2x_2 + \cdots + a_nx_n$.

Then $y \sim N\left(\sum_{i=1}^{n} a_i\,\mu_i,\ \sum_{i=1}^{n} a_i^2\,\sigma_i^2\right)$

EXAMPLE 7-8

A linkage consists of four components as shown in Fig. 7-17. The lengths of x_1, x_2, x_3, and x_4 are known to be $x_1 \sim N(2.0,0.0004)$, $x_2 \sim N(4.5,0.0009)$, $x_3 \sim N(3.0,0.0004)$, and $x_4 \sim N(2.5,0.0001)$. The lengths of the components can be assumed independent, because they are produced on different machines. All lengths are in inches.

The design specifications on the length of the assembled linkage are 12.00 ± 0.10. To find the fraction of linkages that fall within these specification limits, note that y is normally distributed with mean

$$\mu_y = 2.0 + 4.5 + 3.0 + 2.5 = 12.0$$

and variance

$$\sigma_y^2 = 0.0004 + 0.0009 + 0.0004 + 0.0001 = 0.0018$$

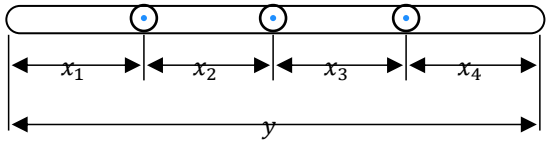


Figure 7-17 A linkage assembly with four components.

$$P\{11.90 \le y \le 12.10\} = P\{y \le 12.10\} - P\{y \le 11.90\}$$
$$= \Phi\left(\frac{12.10 - 12.00}{\sqrt{0.0018}}\right) - \Phi\left(\frac{11.90 - 12.00}{\sqrt{0.0018}}\right)$$
$$= \Phi(2.36) - \Phi(-2.36)$$
$$= 0.99086 - 0.00914$$
$$= 0.98172$$

Nonlinear Combinations

$$y = g(x_1, x_2, \ldots, x_n)$$

$g(\bullet)$: nonlinear function of $x_1, x_2, \ldots, x_n$

μ_i: nominal (i.e., average) dimension for x_i (for i = 1, 2, …, n)

Taylor series expansion of $g(\bullet)$:

$$\rightarrow \quad y = g(\mu_1, \mu_2, \ldots, \mu_n) + \sum_{i=1}^{n} (x_i - \mu_i) \left.\frac{\partial g}{\partial x_i}\right|_{\mu_1, \mu_2, \ldots, \mu_n} + R$$

R is higher order (2 or higher) remainder of the expansion.
$R \rightarrow 0$

$$\rightarrow \quad \mu_y \simeq g(\mu_1, \mu_2, \ldots, \mu_n)$$

$$\sigma_y^2 \simeq \sum_{i=1}^{n} \left(\left.\frac{\partial g}{\partial x_i}\right|_{\mu_1, \mu_2, \ldots, \mu_n} \right)^2 \sigma_i^2$$

EXAMPLE 7-11

Consider the simple DC circuit components shown in Fig. 7-20. Suppose that the voltage across the points (a, b) is required to be 100 ± 2 V. The specifications on the current and the resistance in the circuit are shown in Fig. 7-20. We assume that the component random variables I and R are normally and independently distributed with means equal to their nominal values.

From Ohm's law, we know that the voltage is

$$V = IR$$

Since this involves a nonlinear combination, we expand V in a Taylor series about mean current μ_I and mean resistance μ_R, yielding

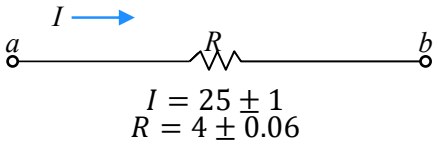


Figure 7-20 Electrical circuit for Example 7-11.

$$\mu_I\mu_R + (I - \mu_I)\mu_R + (R - \mu_R)\mu_I$$

$$\mu_V \simeq \mu_I\mu_R$$

$$\sigma_V^2 \simeq \mu_R^2\sigma_I^2 + \mu_I^2\sigma_R^2$$

Assume I and R are centered at the nominal values.
a = 0.0027: fraction of values falling outside the natural tolerance limits.
Specification limits are equal to natural tolerance limits.

Assume I = 25 ± 1 Amp or $24 \le I \le 26$ and $I \sim N(25, \sigma_I^2)$.

$$Z_{\alpha/2} = Z_{0.00135} = 3.0$$

$$\rightarrow \frac{26 - 25}{\sigma_I} = 3.0 \qquad \rightarrow \sigma_I = 0.33$$

Assume R = 4 ± 0.06 ohms or $3.94 \le R \le 4.06$ and $R \sim N(4, \sigma_R^2)$.

$$\rightarrow \frac{4.06 - 4.00}{\sigma_R} = 3.0 \qquad \rightarrow \sigma_R = 0.02$$

Assuming V is approximately normal:

$$\mu_V \simeq \mu_I \mu_R = (25)(4) = 100 \text{ V}$$

$$\sigma_V^2 \simeq \mu_R^2 \sigma_I^2 + \mu_I^2 \sigma_R^2 = (4)^2(0.33)^2 + (25)^2(0.02)^2 = 1.99 \rightarrow \sigma_V = \sqrt{1.99} = 1.41$$

$$P\{98 \le V \le 102\} = P\{V \le 102\} - P\{V \le 98\} = \Phi\left(\frac{102 - 100}{1.41}\right) - \Phi\left(\frac{98 - 100}{1.41}\right)$$

$$= \Phi(1.42) - \Phi(-1.42) \qquad = \Phi(1.42) - \Phi(-1.42)$$

$$= \Phi(1.42) - \Phi(-1.42)$$

Estimating Natural Tolerance Limits

For normal distribution with unknown mean and variance:

$$\bar{x} \pm Z_{\alpha/2} S \qquad (7\text{-}45)$$

- Difference between tolerance limits and confidence limits
- Nonparametric tolerance limits can also be calculated.


Thank You!